

WEATHER INTEGRATED INTELLIGENT POWER DEMAND FORECASTING INTERFACE



A PROJECT REPORT

in partial fulfilment for the award of the degree

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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Declaration

We solemnly affirm that the Major Project entitled “**Weather Integrated Intelligent Power Demand Forecasting Interface**”, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is our original work carried out under the supervision of Dr. Isha.

We further declare that this work has not been submitted, either in part or in full, to any other University or Institution for the award of any degree or diploma, and that all sources of information and references used in the preparation of this report have been duly acknowledged.

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Certificate

This is to certify that the Major Project entitled “**Weather Integrated Intelligent Power Demand Forecasting Interface**” is a bona fide record of work carried out by

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The project has been completed under my guidance and direct supervision during the academic year 2025–2026. To the best of my knowledge, this work has not been submitted elsewhere for the award of any other degree or diploma.

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Abstract

This report presents the design and development of a data-driven system for electricity demand forecasting and shortage prediction across Indian states. The system integrates historical electricity demand data with meteorological variables such as temperature, humidity, and rainfall to model the influence of climatic conditions on energy consumption patterns.

A structured data preprocessing pipeline and feature engineering techniques, including lag variables and rolling statistics, are employed to capture temporal dependencies effectively.

Multiple machine learning models are evaluated for demand prediction, achieving high predictive accuracy, while a two-stage approach is adopted for shortage prediction, separating occurrence detection and magnitude estimation.

The system is implemented using a scalable backend architecture and a responsive web interface, enabling real-time prediction and user interaction. Additionally, an interactive analytics dashboard provides insights into historical trends, correlations, and system behaviour.

The results highlight strong temporal and regional variability in electricity demand, supporting the need for state-level modeling approaches.

The developed system demonstrates the practical applicability of integrating data analytics, machine learning, and visualization into a unified framework, contributing to improved decision-making in energy planning and grid management.

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List of Abbreviations

- **API** — Application Programming Interface
- **ARIMA** — AutoRegressive Integrated Moving Average
- **CCC** — Cophenetic Correlation Coefficient
- **CNN** — Convolutional Neural Network
- **CSV** — Comma-Separated Values
- **CV** — Cross-Validation
- **CWI** — Composite Weather Index
- **DAX** — Data Analysis Expressions
- **ETL** — Extract Transform Load
- **JSON** — JavaScript Object Notation
- **KPI** — Key Performance Indicator
- **LR** — Linear Regression
- **LSTM** — Long Short-Term Memory
- **MAE** — Mean Absolute Error
- **ML** — Machine Learning
- **MU** — Million Units
- **MW** — Megawatt
- **ORM** — Object-Relational Mapping
- **PSP** — Power Supply Position
- **R²** — Coefficient of Determination
- **RMSE** — Root Mean Square Error
- **ROC-AUC** — Receiver Operating Characteristic - Area Under Curve
- **SCADA** — Supervisory Control and Data Acquisition
- **SVR** — Support Vector Regression
- **UI** — User Interface
- **XGBoost** — Extreme Gradient Boosting

Chapter 1 – Introduction

1.1 Background and Motivation

India’s electricity demand is increasingly influenced by climatic volatility, renewable penetration, urbanization, and regional consumption asymmetry. Traditional forecasting frameworks, primarily based on linear time-series extrapolation, often underperform under nonlinear weather stress and inter-state heterogeneity.

Modern grid analytics requires integrated forecasting systems that combine climatic quantification, machine learning, and deployment-ready infrastructure. Forecasting is no longer limited to statistical accuracy; it must support interpretability, scalability, and operational integration.

In response, this Major Project proposes *Weather-Integrated Intelligent Power Demand Forecasting Interface*, a modular architecture integrating automated data engineering, composite weather modeling, clustering-based segmentation, ensemble forecasting, and API-driven deployment.

The system extends demand forecasting by incorporating shortage prediction, real-time analytical visualization, and an interactive user interface, enabling end-to-end decision support for electricity demand management.

1.2 Problem Statement

Electricity demand at the state-level is influenced by historical consumption patterns, climatic stress, seasonal cycles, and structural demand characteristics. Let daily electricity demand for state s at time t be denoted as $D_{s,t}$.

$$\hat{D}_{s,t+h} = f(D_{s,t-k:t}, W_{s,t-k:t}, X_{s,t}) \quad (1.1)$$

where:

- h is forecast horizon
- k represents lag depth
- $W_{s,t}$ denotes weather variables
- $X_{s,t}$ represents engineered temporal features

In addition to demand, electricity systems must also account for supply-demand imbalances. Let shortage be represented as:

$$\hat{S}_{s,t+h} = g(X_{s,t}) \quad (1.2)$$

where $\hat{S}_{s,t+h}$ denotes predicted shortage.

Existing models often depend primarily on temperature alone, ignore composite climatic stress, and overlook shortage forecasting. Missing weather records and non-state operational entries further degrade model reliability. The core research and engineering problem addressed in this project is:

“How can weather variables be systematically combined with grid demand data to model, analyze, and forecast electricity consumption and shortage across Indian states using data-driven methods, while enabling deployment through an interactive and scalable system?”

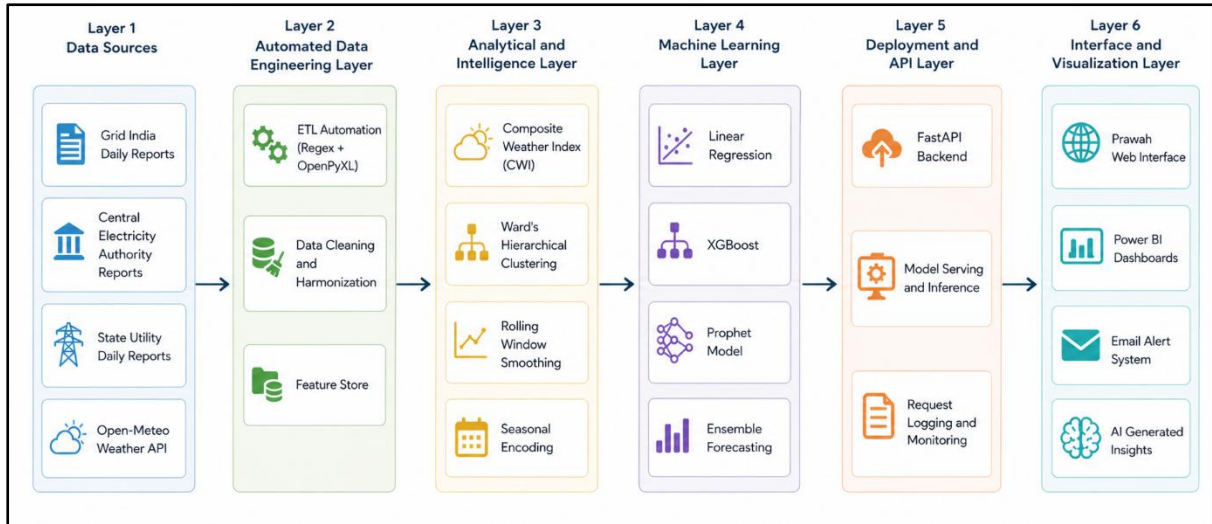


Figure 1.1: Weather-Integrated Intelligent Power Demand Forecasting System

1.3 Aim and Objectives

To architect and implement Prawah (the proposed system), a scalable, weather-integrated intelligent power demand forecasting system that combines advanced analytics, predictive modeling, and full-stack deployment infrastructure. The project objectives are structured across five dimensions: data engineering, analytics, modeling, system architecture, and intelligent interface design.

1.3.1 Data Engineering Objectives

- Integrate multi-source grid datasets (Grid India, CEA, state utilities).
- Develop automated ETL workflows using structured parsing mechanisms.
- Implement state-aware normalization and feature store architecture.
- Enable coordinate-weighted aggregation of meteorological stations.
- Establish scheduled dataset refresh capability.

1.3.2 Analytical Objectives

- Construct a Weather Index (CWI) to quantify multi-dimensional climatic stress.
- Implement Ward's hierarchical clustering for regional segmentation.
- Perform shortage intensity and reliability characterization.
- Conduct state-level elasticity and seasonal decomposition analysis.

1.3.3 Machine Learning Objectives

- Engineer lag-based and rolling-window features.
- Apply cyclical seasonal encoding techniques.
- Train and evaluate multiple models including Linear Regression, XGBoost, Prophet, and ensemble configurations.
- Forecast both maximum demand and shortage metrics.
- Assess robustness under extreme weather conditions.

1.3.4 System Architecture Objectives

- Develop FastAPI-based backend services for model inference.

- Implement RESTful endpoints with structured validation.
- Integrate SQL-based logging and model metadata storage.
- Enable scalable deployment of predictive services.
- Support integration with visualization platforms for analytical reporting.

1.3.5 Intelligent Interface Objectives

- Design an authenticated web interface for user interaction.
- Enable role-based access for different user categories.
- Integrate interactive dashboards for demand prediction and shortage analysis.
- Develop AI-assisted narrative insights for interpretability.
- Provide real-time visualization of forecasts and analytical metrics.

1.4 Scope of the Project

The project focuses on the period from 1 April 2023 to 6 October 2025, covering all major Indian states. The scope of this Major Project encompasses the design and implementation of:

- Automated data acquisition and harmonization pipelines.
- Composite climatic stress modelling.
- Cluster-based state segmentation.
- Multi-model forecasting architecture.
- Shortage prediction framework.
- API-based inference system.
- Authenticated interactive web interface.
- Power BI-based analytical dashboards.
- AI-assisted insight generation modules.

The system operates at state-level granularity across India and is designed for extensibility into broader grid analytics applications. The scope excludes real-time SCADA integration, consumer-level analytics, and national dispatch optimization modelling.

1.5 Significance and Relevance

This project contributes at the intersection of:

- Climate-aware energy analytics
- Advanced machine learning for infrastructure systems.
- Scalable backend engineering
- Cloud-based data visualization
- AI-augmented decision support

It presents a modular framework scalable to decision-support systems for grid operators, analysts, and policymakers, integrating forecasting, shortage analysis, and interactive visualization in a unified platform.

Chapter 2 – Related Work

2.1 Overview: Forecasting Paradigms and Taxonomy

Electricity demand forecasting has evolved significantly over the last three decades, transitioning from purely statistical time-series approaches toward hybrid and intelligent machine learning frameworks. *Hippert et al.* [1] provided one of the earliest comprehensive reviews of neural network applications in load forecasting, highlighting the limitations of linear statistical models in capturing nonlinear weather-demand relationships.

Taylor and Letham [2] introduced scalable time-series forecasting frameworks emphasizing automated seasonality detection and additive decomposition, which later influenced operational forecasting tools. *Chen and Guestrin* [3] proposed XGBoost, a scalable tree boosting framework, which significantly improved regression performance in high-dimensional forecasting tasks and has since been widely adopted in power system analytics.

Recent literature indicates a growing shift toward data-driven learning systems capable of integrating exogenous variables such as temperature, humidity, rainfall, and solar irradiance. *Al-Musaylh et al.* [4] emphasized the comparative superiority of machine learning models over classical regression techniques in volatile load environments. *Kong et al.* [5] demonstrated that Long Short-Term Memory (LSTM) networks outperform conventional regression models in sequential demand prediction.

2.2 Influence of Weather and Renewable Integration

Weather variables have consistently been identified as dominant drivers of electricity demand variability. *Pardo et al.* [6] examined temperature elasticity in European electricity systems and observed nonlinear demand responses during extreme climatic conditions. *Rahman et al.* [7] reported strong state-specific climatic influence patterns in developing economies, reinforcing the need for regionalized modeling.

In the context of renewable integration, solar forecasting has gained particular importance. *Isha et al.* [8] presented a comprehensive review titled “*Solar Power Forecasting: A Review*”, categorizing forecasting methodologies into deterministic, statistical, and artificial intelligence-based models.

The study highlighted the necessity of hybrid modeling approaches for renewable-dominant grids and emphasized the integration of meteorological uncertainty into predictive systems.

Similarly, *Sharma et al.* [9] investigated solar irradiance forecasting and noted that combining physical and machine learning models enhances grid stability forecasting. *Zhou et al.* [10] extended this concept to integrated energy systems, demonstrating the importance of multi-variable climatic inputs.

2.3 Machine Learning and Hybrid Forecasting Models

Machine learning techniques have increasingly dominated load forecasting research. Random Forest and Support Vector Regression models have demonstrated robustness under noisy conditions, as reported by *Ahmad et al.* [11]. XGBoost-based forecasting models were shown by *Li et al.* [12] to improve prediction stability in state-level load datasets.

Deep learning architectures have further advanced predictive capabilities. *Guo et al.* [13] developed a CNN-LSTM hybrid architecture capable of capturing both spatial and temporal dependencies. *Mu et al.* [14] proposed a Seq2Seq LSTM approach for multi-step forecasting with improved generalization.

Hybridization strategies are now considered best practice. *Osório et al.* [15] implemented ensemble blending models combining statistical and machine learning frameworks to enhance forecasting robustness. *Arslan et al.* [16] demonstrated that combining Prophet with LSTM architectures significantly reduced forecasting errors compared to standalone models.

2.4 Regionalization and Clustering-Based Approaches

National aggregation often conceals regional heterogeneity in demand elasticity. *Fan et al.* [17] observed that regional clustering improves model generalization in heterogeneous load zones. *Zengxi* [18] implemented attention-based clustering for load segmentation and achieved improved prediction accuracy.

Ward's hierarchical clustering method has been applied in various energy analytics studies to group similar consumption patterns. *Khaldi et al.* [19] demonstrated that clustering prior to modeling reduces variance in regional forecasting models. *Yona et al.* [20] also emphasized that cluster-driven forecasting frameworks provide more operationally meaningful insights than centralized models.

Feature engineering remains critical in time-series forecasting. Rolling window smoothing techniques were evaluated by *Abumohsen* [21], demonstrating improved stability in demand modeling. Composite weather indices have been proposed to aggregate multi-dimensional meteorological stress. *Chitalia et al.* [22] suggested normalized weather aggregations for load elasticity studies in India.

2.5 Deployment-Oriented Forecasting Systems

While predictive accuracy has improved significantly, operational deployment remains underexplored in academic literature. *Chen and Guestrin* [3] discussed scalable model deployment implications of gradient boosting systems. *Taylor and Letham* [2] emphasized production-ready forecasting pipelines. *Fidalgo et al.* [23] presented web-based energy forecasting dashboards integrated with API-driven backends.

Kumar and Jain [24] demonstrated Power BI-based grid analytics systems with automated refresh cycles. *Sivaraman et al.* [25] proposed geospatial visualization frameworks for state-wise electricity demand monitoring. Containerized deployment using Docker has become standard practice in ML system engineering, as outlined by *Merkel* [26]. FastAPI-based microservices have been widely adopted for low-latency forecasting APIs [27].

2.6 Identified Research Gap

Although extensive work exists in demand forecasting, most studies either focus on predictive accuracy or analytical evaluation in isolation. Limited research integrates multi-variable weather modeling, shortage prediction, clustering-based segmentation, and deployment-ready system architecture within a unified framework. This gap highlights the need for an end-to-end system that combines analytics, modeling, and operational deployment.

Chapter 3 – Research Methodology

3.1 Methodological Framework

The proposed system, ‘Prawah’, follows a layered analytical and engineering architecture integrating data harmonization, climatic stress quantification, predictive modeling, and deployment engineering. The methodological pipeline is structured to transform heterogeneous grid and meteorological data into interpretable and deployable forecasting intelligence. The framework proceeds sequentially as:

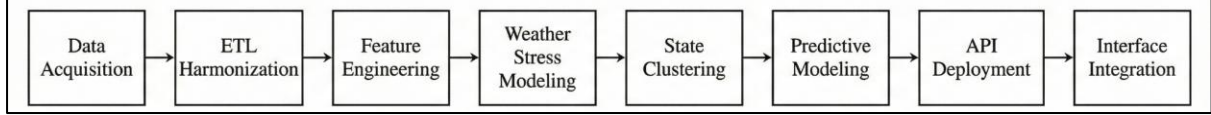


Figure 3.1: Research Methodology Pipeline

3.2 Data Acquisition and Harmonization

The dataset integrates daily power system reports and meteorological records. Power system variables include: Maximum Demand Met, Energy Met and Shortage. Weather variables include temperature, humidity, and rainfall.

Average temperature is computed as:

$$T_{avg,s,t} = \frac{T_{s,t}^{max} + T_{s,t}^{min}}{2} \quad (3.1)$$

State-level weather aggregation is performed using weighted spatial averaging:

$$W_{s,t} = \sum_{i=1}^{n_s} w_i W_{i,t}, \text{ where } \sum w_i = 1 \quad (3.2)$$

Automated ETL processes ensure schema normalization, missing value handling, and structured dataset creation suitable for modeling.

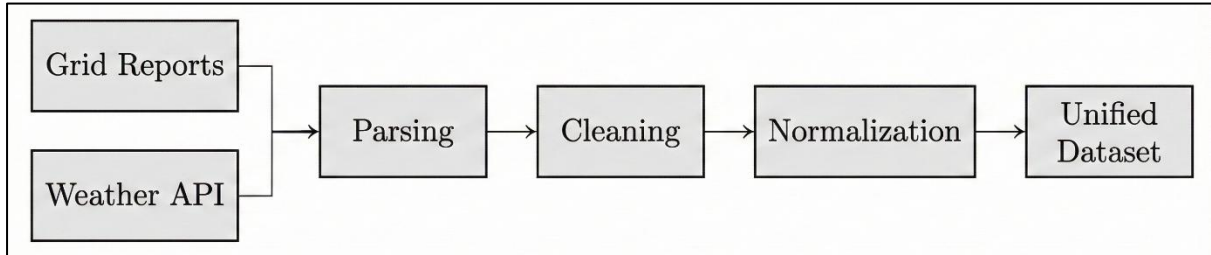


Figure 3.2: Data Acquisition and Harmonization Framework

3.3 Feature Engineering and Climatic Stress Modeling

Temporal dynamics are encoded using lag and smoothing mechanisms.

- Lag feature:

$$L_{s,t}^{(k)} = D_{s,t-k} \quad (3.3)$$

- Rolling mean smoothing:

$$\bar{D}_{s,t} = \frac{1}{m} \sum_{i=0}^{m-1} D_{s,t-i} \quad (3.4)$$

- Cyclical encoding preserves seasonality:

$$Month_{sin} = \sin\left(\frac{2\pi M_t}{12}\right) \quad (3.5)$$

$$Month_{cos} = \cos\left(\frac{2\pi M_t}{12}\right) \quad (3.6)$$

To quantify multi-dimensional weather impact, a Composite Weather Index (CWI) is introduced. After normalization:

$$Z_x = \frac{x - \mu_x}{\sigma_x} \quad (3.7)$$

$$CWI_{s,t} = \frac{Z_T + Z_H + Z_R}{3} \quad (3.8)$$

This reduces heterogeneous climatic variables into a unified stress scalar.

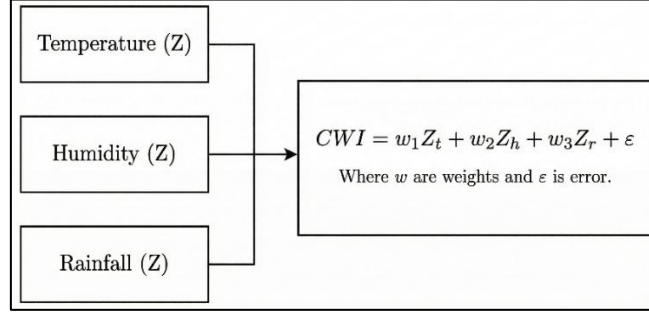


Figure 3.3: Composite Weather Index Formulation

3.4 State Segmentation via Clustering

Inter-state heterogeneity is addressed through hierarchical clustering. Each state is represented by a feature vector:

$$F_s = [\rho_T, \rho_H, \rho_R, SeasonalAmplitude_s, ShortageRates_s] \quad (3.9)$$

Ward's method minimizes within-cluster variance:

$$\Delta E = \sum_{i \in C} (x_i - \bar{x})^2 \quad (3.10)$$

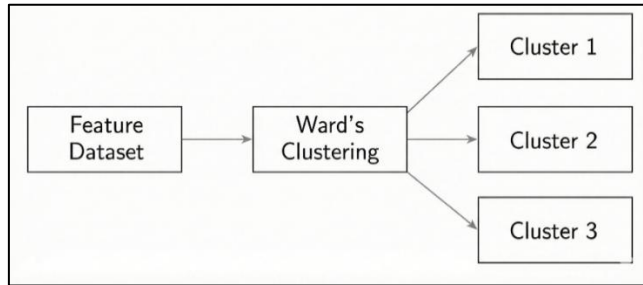


Figure 3.4: State Clustering Framework

3.5 Predictive Modeling Framework

Multiple models are implemented to capture complementary characteristics.

- Linear regression baseline:

$$\hat{D}_{s,t} = \beta_0 + \sum_{i=1}^n \beta_i X_{i,s,t} \quad (3.11)$$

- Gradient boosting formulation:

$$\hat{D}_{s,t} = \sum_{m=1}^M \gamma_m h_m(X_{s,t}) \quad (3.12)$$

- Prophet additive decomposition:

$$D_{s,t} = g(t) + s(t) + h(t) + \epsilon_t \quad (3.13)$$

- Ensemble prediction:

$$\hat{D}_{ensemble} = \alpha_1 \hat{D}_{LR} + \alpha_2 \hat{D}_{XGB} + \alpha_3 \hat{D}_{Prophet} \quad (3.14)$$

subject to: $\sum \alpha_i = 1$

Model validation minimizes RMSE described in Equation 6.14.

3.6 Deployment and System Integration

The forecasting models are deployed through scalable web-based infrastructure. Model inference is exposed through API endpoints:

$$API(State, Date, Weather) \rightarrow \hat{D}_{s,t}, \hat{S}_{s,t} \quad (3.15)$$

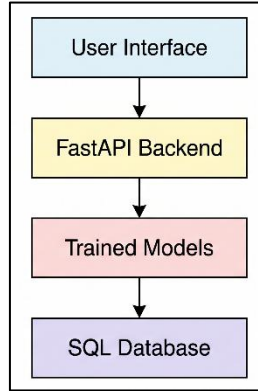


Figure 3.5: Deployment and API Architecture

3.7 System Architecture

The complete architecture of Prawah follows a layered, modular design enabling transformation from heterogeneous data sources to deployment-ready forecasting intelligence. The system integrates data ingestion, automated harmonization, climatic stress modeling, ensemble forecasting, deployment infrastructure, and visualization within a unified pipeline.

The deployment layer integrates Render-hosted APIs and Vercel-hosted user interface maintaining the same modular architecture principles.

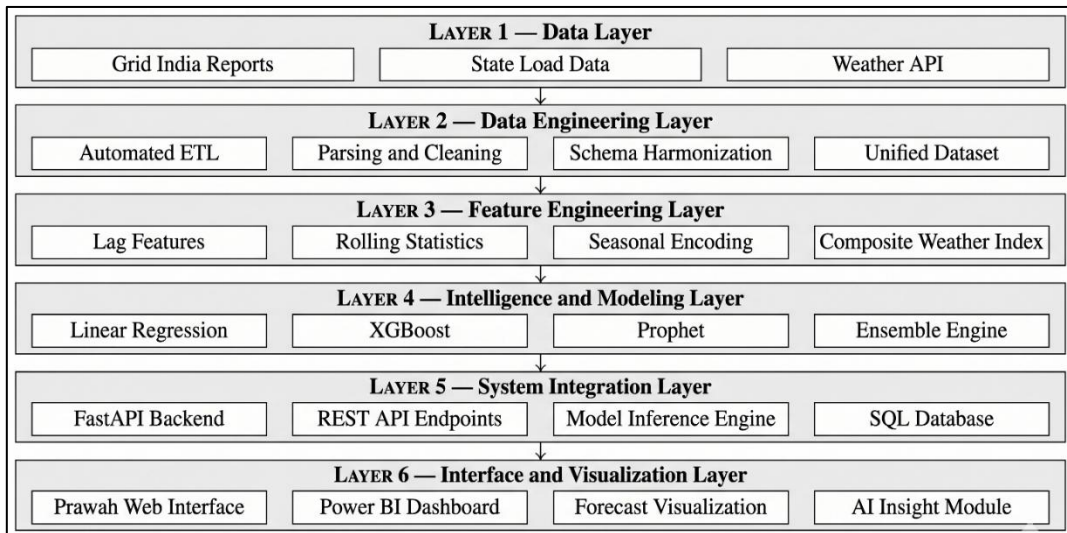


Figure 3.6: Complete System Architecture

Chapter 4 – Data Engineering and Preprocessing

4.1 Data Sources and Collection Strategy

Two primary data streams were integrated.

- Power System Operational Data: Daily Power Supply Position (PSP) reports obtained from Grid India for fiscal years 2023–24, 2024–25, and 2025–26 (partial).
- Meteorological Data: Historical weather observations retrieved from the Open-Meteo Archive API covering 1 April 2023 to 6 October 2025.

For each state, three to four representative geographic coordinates were selected to capture climatic diversity. State-level weather values were computed using coordinate-weighted averaging to ensure balanced representation.

TABLE 4.1: GRID (ELECTRICITY) DATA ACQUISITION DETAILS

File Name	Rows	Columns	Coverage Period
Combined_PSP_Data202324.xlsx	13,907	12	Apr 2023 – Mar 2024
Combined_PSP_Data202425.xlsx	13,874	12	Apr 2024 – Mar 2025
Combined_PSP_Data202526.xlsx	7,371	12	Apr 2025 – Oct 2025
Calendar_Progression_PSP.xlsx	35,152	12	Apr 2023 – Oct 2025

TABLE 4.2: WEATHER DATA ACQUISITION DETAILS

File	India_Statewise_Weather_2023_2025.csv
Rows	33,121
Columns	State, Date, Year, Month, Day, Temp_Max, Temp_Min, Humidity, Rainfall
Date Range	1 April 2023 – 6 October 2025
Coverage	36 State entities according to Grid India taxonomy
Computation	State-level averages derived via Coordinate-Weighted Mean

4.2 Data Integration and Assembly

The raw datasets are expected to exhibit variations in naming conventions, formats, and timestamp alignment. A structured ETL pipeline will be implemented consisting of:

- State name normalization using a standardized lookup mapping
- Aggregation of non-state operational entities into a consolidated category
- Left-join merging of electricity and weather datasets on (State, Date)
- Computation of derived variables like Avg_Temp, ShortageCalculated_MW, shortage_flag and temporal markers.

4.3 ETL Workflow

The proposed workflow of Extract-Transform-Load Pipeline is described in Figure 4.1.

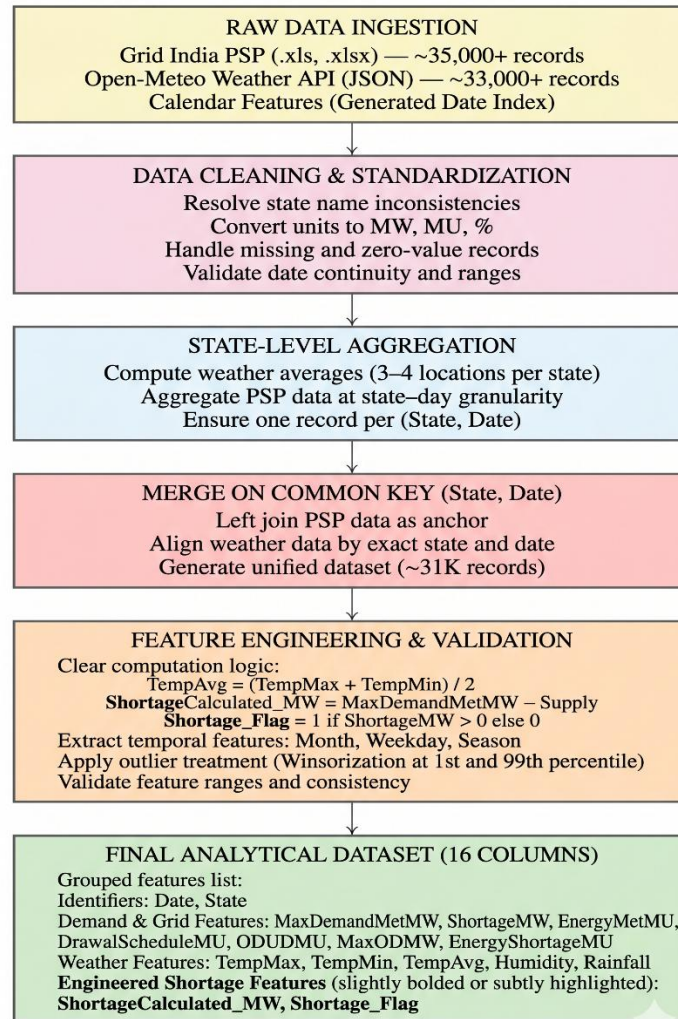


Figure 4.1: ETL Flow Diagram

4.4 Data Quality Handling

The preprocessing stage will address anticipated data quality issues including zero-valued demand entries due to missing reports, state name inconsistencies across sources, missing weather records and non-state operational entries.

TABLE 4.3: DATA QUALITY CORRECTIONS APPLIED

Correction Types	Count	Method Applied	Impact
Zero-valued (Date, State) records	1,443	Forward-backward interpolation per state	Removed synthetic demand dips; preserved trend
State name normalization	~200 variants	Standardized to title-case lookup	Ensured proper merging of state-wise records
Missing-report handling	Embedded in 1,443	Per-state ffill/bfill	Continuous time series per state
Extra non-state entries	~5 (Railways, RIL Jamnagar etc.)	Aggregated into "Other" category	Maintained national electricity totals

4.5 Analytical Dataset Schema

The final integrated dataset is expected to consist of 16 standardized columns combining electricity, weather, and temporal information. Each row will correspond to one state on one calendar day.

TABLE 4.4: DATASET SCHEMA

#	Column	Unit	Source	Description
1	Date	YYYY-MM-DD	Grid & Weather	Calendar day
2	State	—	Grid & Weather	State identifier
3	MaxDemandMetMW	MW	Grid India	Peak demand met
4	ShortageMW	MW	Grid India	Load shortfall
5	EnergyMetMU	MU	Grid India	Total delivered energy
6	DrawalScheduleMU	MU	Grid India	Scheduled transfer
7	ODUDMU	MU	Grid India	Over/under drawal
8	MaxODMW	MW	Grid India	Max deviation
9	EnergyShortageMU	MU	Grid India	Energy deficit
10	TempMax	°C	Open-Meteo	Daily high temperature
11	TempMin	°C	Open-Meteo	Daily low temperature
12	TempAvg	°C	Computed	(TempMax + TempMin)/2
13	Humidity	%	Open-Meteo	Relative humidity
14	Rainfall	mm	Open-Meteo	Daily precipitation
15	ShortageCalculated_MW	MW	Computed	Derived supply-demand gap
16	shortage_flag	Binary (0/1)	Computed	Shortage Indicator

4.6 Outlier Handling and Feature Refinement

To mitigate distortion from rare anomalies, Winsorization will be applied at the 1st and 99th percentiles on a per-state basis. Temporal features including year, month, day-of-week, and seasonal classification will be derived to facilitate structured analytics. Season categories will be defined as: Summer (March–May), Monsoon (June–September), Post-Monsoon (October–November) and Winter (December–February). The engineered shortage variables are also validated to ensure consistency with observed demand trends and are incorporated into downstream analytical and modeling workflows.

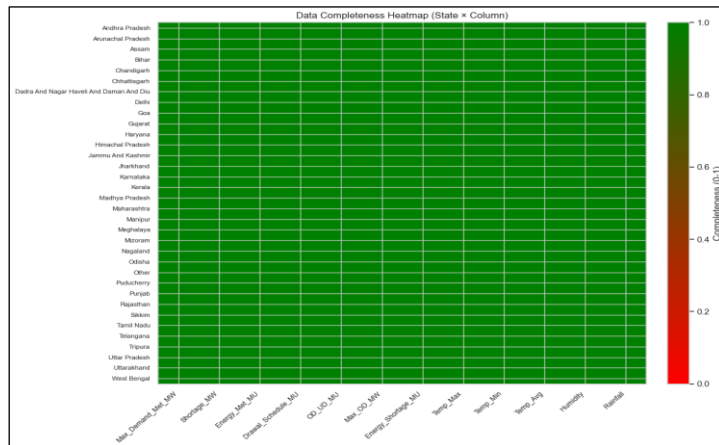


Figure 4.2: Data Completeness Heatmap

Chapter 5 – Analytical Framework

5.1 Descriptive Statistical Profiling

Descriptive statistics, such as mean, median, and standard deviation, were calculated for each state to describe central tendency, dispersion, and daily variability with the objective to establish a solid analytical basis. These characteristics provide the foundation for evaluating weather-related instability and identifying anomalous demand behaviour.

In addition to demand analysis, the framework also incorporates shortage-based analytics using derived variables to better understand supply-demand imbalance across states.

5.2 Correlation Modeling

The Pearson correlation coefficient [8] is used for calculating linear relationships between maximum daily electricity consumption and meteorological variables (temperature, humidity, and rainfall). Two hierarchical levels were used for the analysis:

1) *National Aggregation*: To evaluate the national grid's overall climate reactivity, demand and weather variables were taken as an average across all states.

2) *State-Level Modelling*: To find hidden regional relationships and geographical fluctuations, independent correlation coefficients were calculated for each of 34 state entities.

To ensure the accuracy of the relationships found, statistical significance was evaluated using p-values ($p < 0.05$).

5.2.1 National Correlation

Aggregated correlations at the national level are weak: Temperature Correlation = 0.286, Humidity Correlation = -0.188 , and Rainfall Correlation = -0.113 , all of which are statistically significant with $p < 0.05$. Although these correlations are present, their small sizes imply that local variability and practical significance are hidden by national averages.

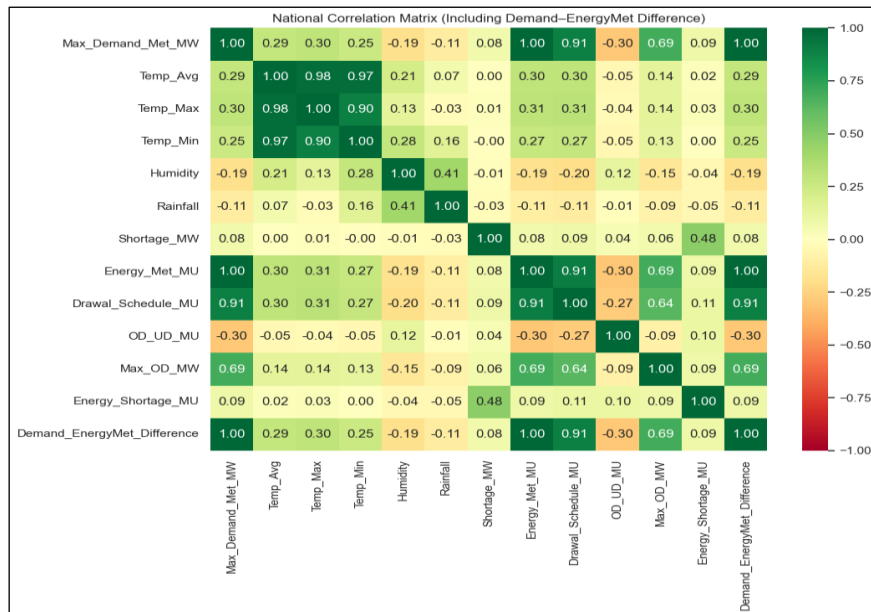


Figure 5.1: National Correlation Matrix

5.2.2 State-Level Weather–Demand Correlation

Per-state correlation coefficients are summarised in Table 5.1. Strong positive relationships ($r > 0.7$) are seen in a number of coastal and eastern states, indicating considerable sensitivity to cooling. Hilly and arid states, on the other hand, show opposite tendencies, probably as a result of changes in industrial or agricultural activities. 19 (or roughly 56%) of the 34 states have $|r| > 0.5$, indicating that significant variation among regions is hidden by national aggregation—an illustration of the “averaging fallacy.”

TABLE 5.1: CORRELATION COEFFICIENTS WITH MAXIMUM DEMAND

State	r_Temp_Avg	r_Humidity	r_Rainfall
Assam	0.885481	0.241874	0.205984
West Bengal	0.866526	0.239709	0.228481
Kerala	0.800974	-0.53186	-0.41444
Odisha	0.752607	-0.09269	0.036571
Puducherry	0.749797	-0.46374	-0.21568
Bihar	0.744321	0.075162	0.225014
Uttar Pradesh	0.72963	0.049699	0.217003
Goa	0.723619	-0.53092	-0.48936
Tamil Nadu	0.701604	-0.63545	-0.34714
Punjab	0.655478	0.149626	0.114246
Delhi	0.641595	0.080877	0.191981
Haryana	0.638249	0.0422	0.171457
Chandigarh	0.574989	0.183031	0.099686
Karnataka	0.416079	-0.79176	-0.49411
Andhra Pradesh	0.347431	-0.50328	-0.45545
Chhattisgarh	0.345566	-0.43084	-0.23243
Nagaland	0.344911	0.362489	0.102249
Tripura	0.332712	0.081907	-0.01024
Gujarat	0.325189	-0.53796	-0.4112
Jharkhand	0.306439	-0.00024	-0.0232
Maharashtra	0.276094	-0.81241	-0.5953
Uttarakhand	0.185273	-0.09544	-0.13309
Arunachal Pradesh	0.134977	-0.07091	-0.27774
Dadra & Nagar Haveli and Daman and Diu	0.031298	0.169773	0.133098
Telangana	-0.13542	-0.26151	-0.26536
Rajasthan	-0.382	-0.26129	-0.44436
Manipur	-0.39093	-0.17337	-0.23712
Meghalaya	-0.5071	-0.31585	-0.21608
Himachal Pradesh	-0.53376	-0.29536	-0.10927
Jammu And Kashmir	-0.57327	-0.16191	-0.0812
Mizoram	-0.57659	-0.27119	-0.33441
Madhya Pradesh	-0.58744	-0.31017	-0.4835
Sikkim	-0.71323	-0.35884	-0.08045

5.3 Rolling Window Smoothing

Operational and behavioural variations make daily demand values inherently noisy. A rolling mean filter was used using 7-day and 14-day frames to extract underlying trends.

While the 14-day mean highlights medium-term seasonal transitions, the 7-day mean reflects short-term weekly oscillations. The smoothed medium-term trends for each of the 34 states are shown simultaneously in Figure 5.2.

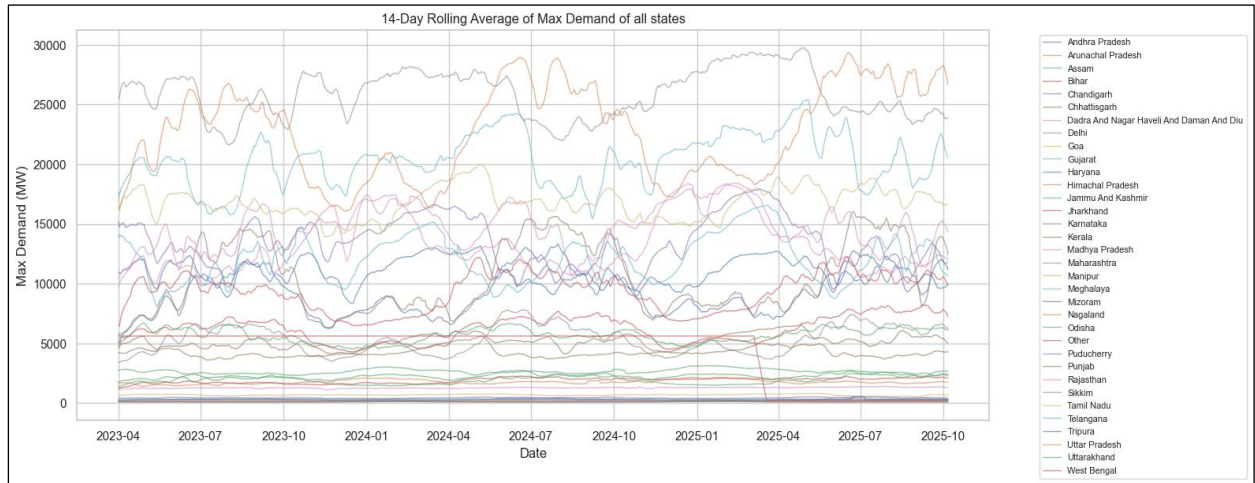


Figure 5.2: 14-Day Rolling Average of Max Demand of All States

5.4 Shortage and Reliability Diagnostics

By monitoring daily supply-demand disparities, operational reliability was evaluated. State and national shortage magnitudes were combined, and persistent deficit periods were identified using a 14-day rolling mean.

With average daily shortfalls of 53 MW and 41 MW, respectively, Bihar and Maharashtra have the largest persistent shortages. The significance of focused planning instead of uniform allocation is shown by these localised difficulties.

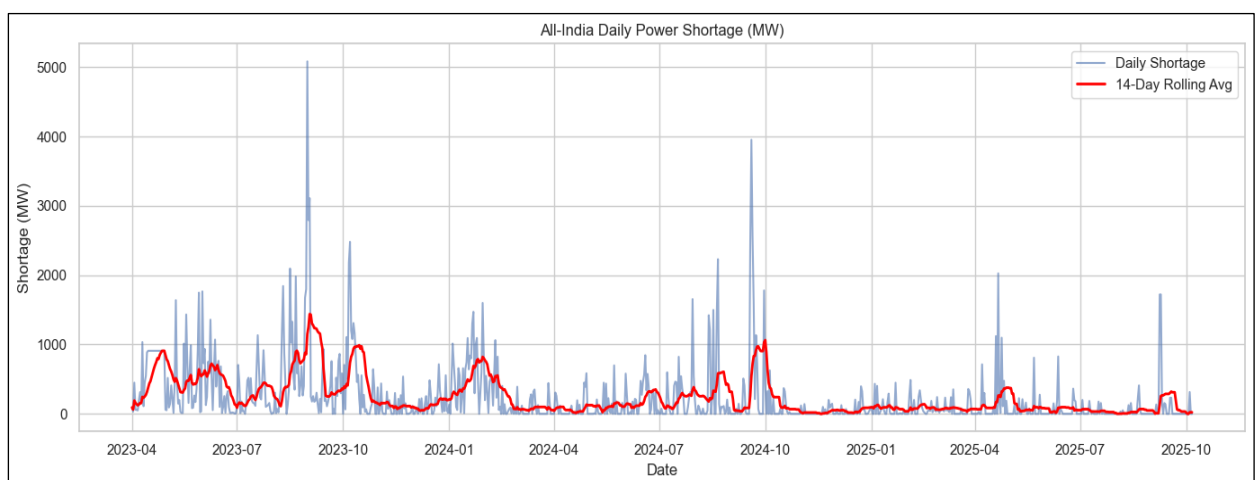


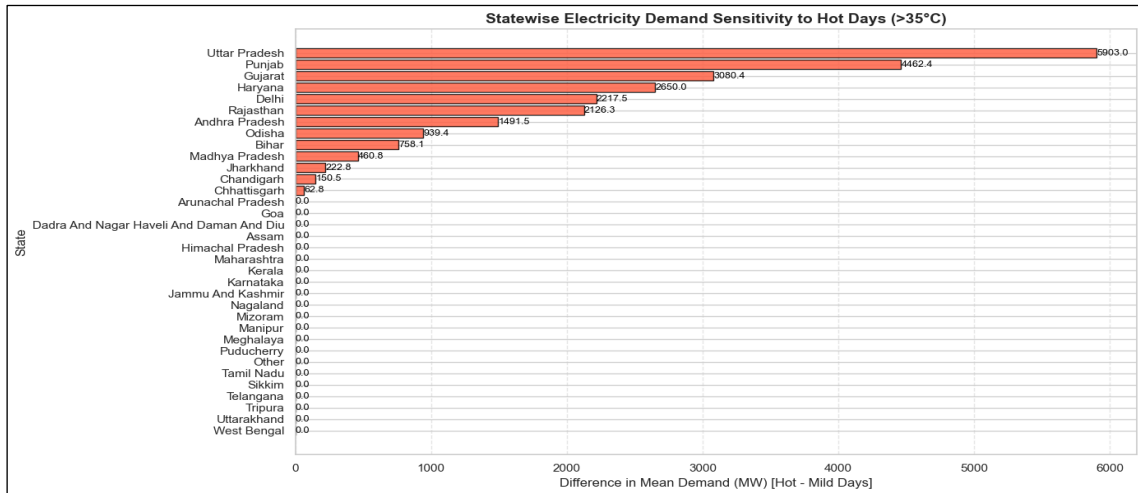
Figure 5.3: All-India Daily Power Shortage (MW)

TABLE 5.2: STATES WITH FREQUENT SHORTAGES

Rank	State	Average Daily Shortage (MW)
1	Bihar	53.046892
2	Maharashtra	41.003272
3	Jammu And Kashmir	26.763359
4	Uttar Pradesh	25.870229
5	Haryana	21.816794
6	Jharkhand	19.283533
7	Madhya Pradesh	9.509269
8	Assam	6.031625

5.5 Temperature Sensitivity Assessment

Every observation was classified as a Hot Day or Mild Day according to a 35 °C threshold in order to investigate direct heat-related demand stress. The difference between the two groups' mean maximum demand was used to quantify the temperature-induced demand premium. Sensitivity to extreme heat conditions can be compared across states using this metric, which is shown in Figure 5.4.

**Figure 5.4: Statewise Electricity Demand Sensitivity to Hot Days (>35 °C)**

5.6 Composite Weather Index (CWI) Formulation

Three standardised weather indicators: temperature, humidity, and rainfall; were integrated to create a composite index. The index is expressed as:

$$CWI = \frac{1}{3}(z_{TempAvg} + z_{Humidity} + z_{Rainfall}) \quad (5.1)$$

$$\text{where } z = \frac{x - \mu}{\sigma} \quad (5.2)$$

For the majority of states, using CWI to integrate temperature, humidity, and rainfall improves the accuracy. While the national correlation $r(CWI) = 0.075$ is still poor, West Bengal ($r = 0.577$), Assam ($r = 0.573$), and Bihar ($r = 0.555$) show substantial multi-factor weather responses. This demonstrates that regional climate-demand linkages are overlooked by aggregate-level statistics.

5.7 Hierarchical Clustering and Validation

To generalize weather–demand behavior across regions, hierarchical clustering was employed on standardized correlation vectors for temperature, humidity, and rainfall. Each state’s feature triplet ($rTemp$, $rHumidity$, $rRainfall$) was scaled using z-normalization and clustered using Ward’s minimum variance method. The resulting dendrogram identified four statistically cohesive groups, confirmed by a cophenetic correlation coefficient (CCC) of 0.802, indicating strong hierarchical consistency. A unique functional and climatic pattern is associated with each cluster.

- 1) *Red* denotes Humidity-Dominant Coastal Regions like Kerala, Goa, etc.
- 2) *Orange* denotes Heat-Sensitive plains such as Bihar, UP, etc.
- 3) *Green* denotes Cold-Sensitive Hills like Meghalaya, Himachal Pradesh, etc.
- 4) *Grey* denotes Industrial-Stable / Mixed Areas such as Jharkhand, Uttarakhand, etc.
- 5) Regions with unavailable data are represented by grey-marked lines like Ladakh.

These clusters are shown spatially throughout the Indian subcontinent in Figure 5.5. The dendrogram shows that functional and climatic similarities are geographically related: coastal and eastern states demonstrate strong humidity-based sensitivity, while northern and hilly regions cluster together due to cold weather reactions.

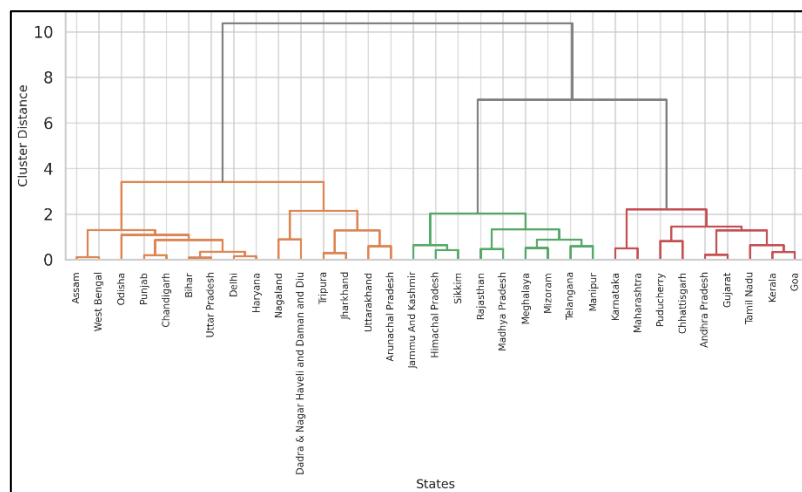


Figure 5.5: Dendrogram of Indian States

5.8 Interactive Dashboard Visualization Strategy

The analytical findings are presented through an interactive Power BI dashboard built using a star schema model, connecting main PSP_Weather_Merged.csv (Fact Table) and aggregated CSVs to Date and State Dimension Tables. There are 5 modules in the dashboards:

- 1) National Overview
- 2) State-wise Insights
- 3) Temperature–Demand Correlation Page
- 4) Shortage & Reliability Page
- 5) Humidity and Rainfall Correlation Page

Chapter 6 – Machine Learning Framework

6.1 Prologue

The machine learning pipeline is designed as a production-grade analytical framework for electricity demand forecasting and shortage prediction. The architecture integrates time-series forecasting, supervised learning, and a structured feature engineering pipeline to ensure robustness, interpretability, and deployment readiness.

The system operates at a state-level granularity and supports both predictive and probabilistic outputs for grid-level decision support.

6.2 Problem Formulation

The modeling framework consists of two primary predictive tasks:

6.2.1 Demand Forecasting

$$\widehat{D}_{s,t+h} = f(X_{s,t}), \forall s \in S, t \in T \quad (6.1)$$

where:

- $\widehat{D}_{s,t+h}$: Predicted demand for state s at horizon h
- $X_{s,t}$: Feature set including weather and temporal variables

6.2.2 Shortage Forecasting

- Classification (Occurrence):

$$\hat{P}(S_{s,t} = 1) = f_c(X_{s,t}) \quad (6.2)$$

- Regression (Magnitude):

$$\hat{S}_{s,t+h} = f_r(X_{s,t}), S_{s,t} > 0 \quad (6.3)$$

This dual formulation ensures separation between event detection and severity estimation, improving model stability.

6.3 Feature Engineering Pipeline

The feature engineering layer is structured to capture temporal dependencies while strictly avoiding data leakage.

Temporal Encoding:

- Day-of-week, quarter, seasonal segmentation
- Weekend indicators
- Cyclical transformations:

$$Month_{sin} = \sin\left(\frac{2\pi \cdot Month}{12}\right) \quad (6.4)$$

$$Month_{cos} = \cos\left(\frac{2\pi \cdot Month}{12}\right) \quad (6.5)$$

$$DOW_{sin} = \sin\left(\frac{2\pi \cdot DayOfWeek}{7}\right) \quad (6.6)$$

$$DOW_{cos} = \cos\left(\frac{2\pi \cdot DayOfWeek}{7}\right) \quad (6.7)$$

Lag-Based Features:

$$Lag_1 = D_{t-1} \quad (6.8)$$

$$Lag_7 = D_{t-7} \quad (6.9)$$

Rolling Features (Leakage-Free):

$$Demand_{7d_avg}(t) = \frac{1}{7} \sum_{i=1}^7 D_{t-i} \quad (6.10)$$

$$Shortage_{7d_avg}(t) = \frac{1}{7} \sum_{i=1}^7 S_{t-i} \quad (6.11)$$

Derived Stress Indicator:

$$Demand_Gap_Ratio = \frac{Shortage_{7d_avg}}{Demand_{7d_avg} + Shortage_{7d_avg}} \quad (6.12)$$

This feature captures system stress using only historical data, ensuring no leakage from the target variable.

6.4 Model Benchmarking and Selection

A structured model comparison framework is implemented using time-series cross-validation to preserve chronological order. Evaluated Models are:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Support Vector Regression (SVR)

Performance evaluation is based on:

$$RMSE = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2} \quad (6.13)$$

$$MAE = \frac{1}{n} \sum |y - \hat{y}| \quad (6.14)$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (6.15)$$

The results demonstrate that SVR fails to capture temporal dynamics effectively and is therefore excluded from the final architecture.

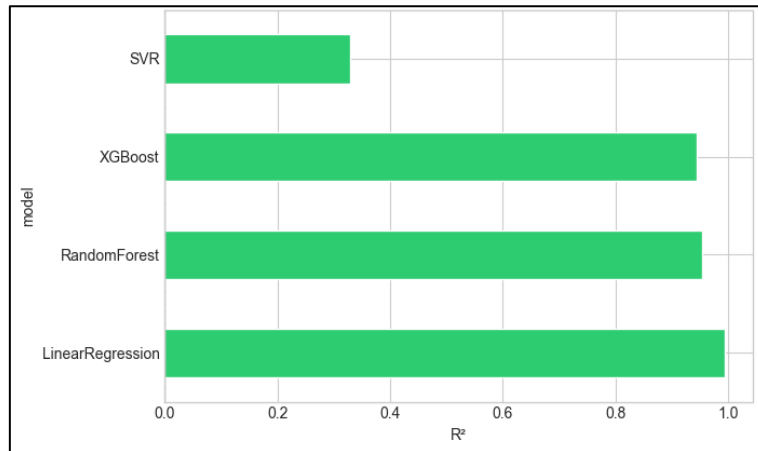


Figure 6.1: Mean R^2 across CV Folds

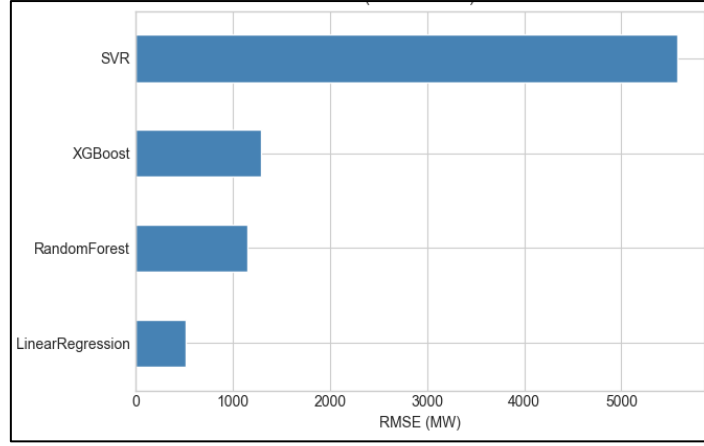


Figure 6.2: Mean RMSE across CV Folds

6.5 Demand Forecasting using Prophet

Demand forecasting is implemented using Prophet, which decomposes the time series into interpretable components:

$$y(t) = g(t) + s(t) + \epsilon_t \quad (6.16)$$

Where:

- $g(t)$: Trend component
- $s(t)$: Seasonal component
- ϵ_t : Residual noise

Key configurations include yearly and weekly seasonality enabled, additive seasonality mode, and increased changepoint sensitivity to capture demand growth trends. Weather regressors are excluded due to instability across smaller states, improving generalization consistency.

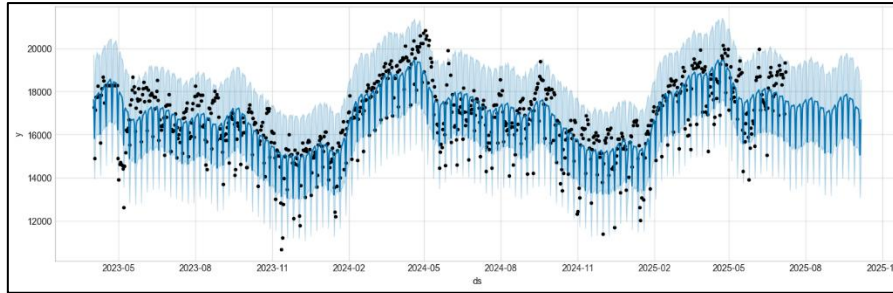


Figure 6.3: Prophet Demand Forecast – Tamil Nadu (shaded = 95% confidence interval)

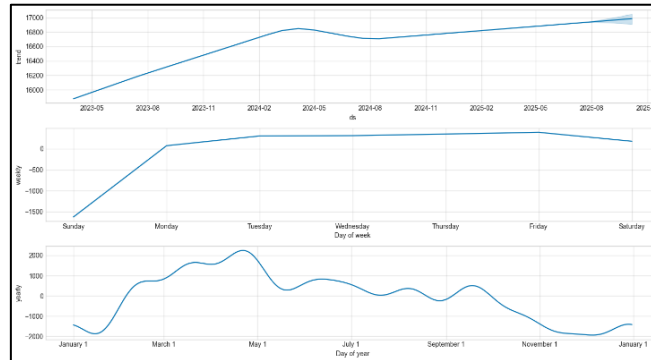


Figure 6.4: Prophet Components - Tamil Nadu

6.6 Shortage Prediction Architecture

6.6.1 Shortage Classification Model

- Algorithm: Gradient Boosting Classifier
- Output: Probability of shortage occurrence

The model is trained using stratified sampling to preserve class distribution and evaluated using ROC-AUC.

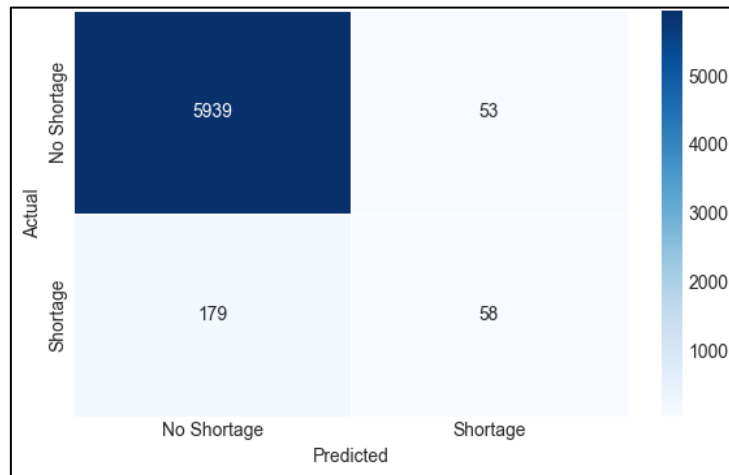


Figure 6.5: Shortage Classifier – Confusion Matrix

6.6.2 Shortage Magnitude Regression Model

- Algorithm: Random Forest Regressor
- Trained only on shortage-positive samples

This design eliminates bias toward zero and improves predictive accuracy for actual shortage events.

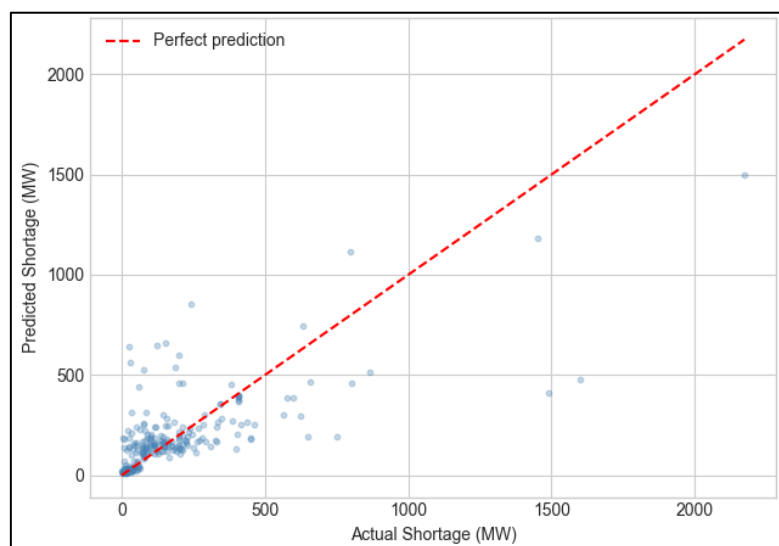


Figure 6.6: Shortage Magnitude – Actual vs Predicted

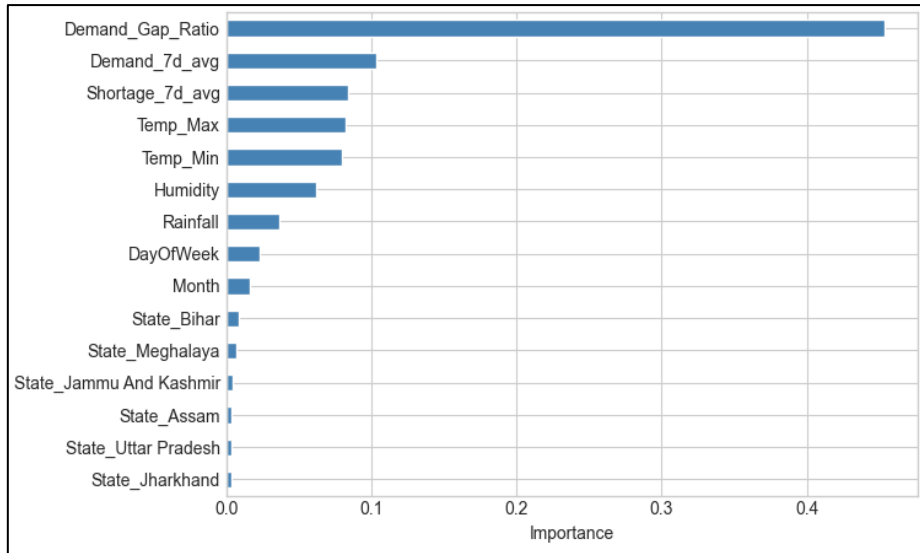


Figure 6.7: Top 15 Feature Importances

6.7 Model Validation Strategy

The validation framework is aligned with time-series constraints:

- TimeSeriesSplit for regression models
- Train-test split with stratification for classification
- Separate evaluation pipelines for each modeling task

This ensures that future information is never used during training, maintaining real-world applicability.

6.8 Model Persistence and Integration

All trained models and metadata are serialized and stored for deployment:

$$\text{Artifacts} = \{\text{Prophet}_s, \text{Classifier}, \text{Regressor}, \text{FeatureSchema}\} \quad (6.17)$$

Artifacts include:

- State-wise Prophet models
- Shortage classifier and regressor
- Feature metadata for inference consistency
- Evaluation summaries

This modular storage structure enables seamless integration with the API layer.

Chapter 7 – Power BI Visualization Interface

7.1 Visualization Overview

The Power BI interface serves as the visualization and decision-support layer of the system. It transforms processed data and analytical outputs into an interactive environment where users can explore electricity demand, climatic relationships, and shortage patterns.

The interface is designed to support dynamic filtering, multi-level aggregation, and automated insight generation, enabling both exploratory and guided analysis.

7.2 Data Model Architecture

The dashboard is built using a *star schema architecture*, ensuring efficient query execution and logical data organization.

- **Fact Table:** PSP_Weather_Merged (state–date level data)
- **Dimension Tables:**
 - DateDimension (date hierarchy: day, month, quarter, year)
 - StateDimension (state-level filtering and grouping)
- **Supporting Tables:**
 - Shortage_Trend (rolling shortage metrics)
 - Statewise_Monthly_Demand (monthly aggregation)
 - Statewise_Seasonal_Demand (seasonal aggregation)
 - Weather_Correlation_By_State (precomputed correlations)
- **Measure Table:** Centralized storage of all DAX measures, including KPIs, statistical metrics, and narrative output.

This structure ensures:

- Efficient cross-filtering across visuals
- Separation of raw data and computed logic
- Scalability for additional analytical layers

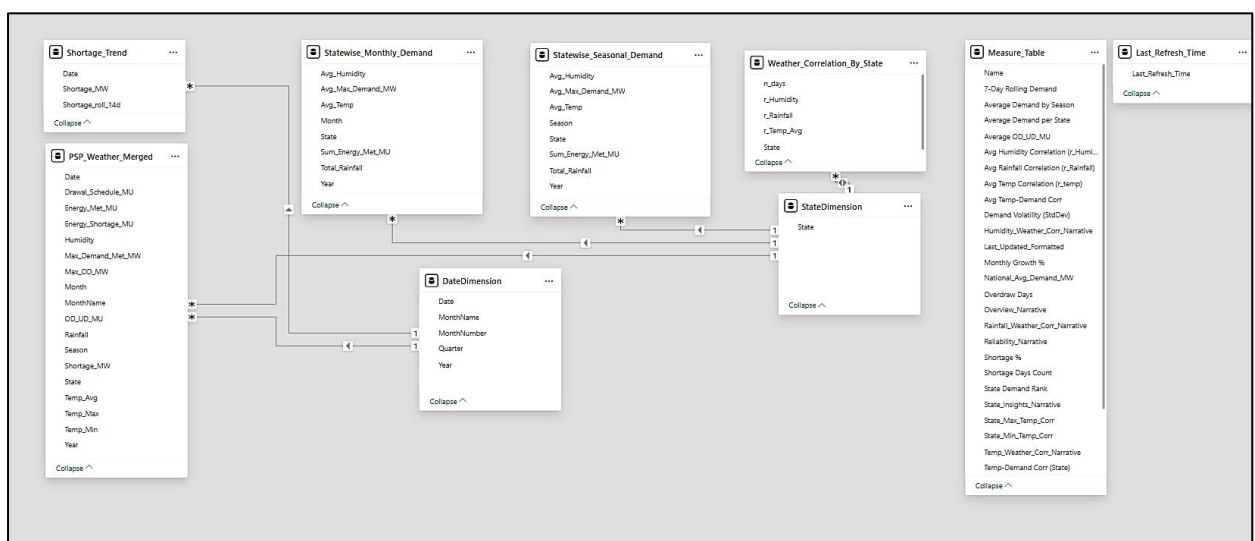


Figure 7.1: Power BI Data Model (Star Schema)

7.3 Dashboard Modules

The dashboard is divided into five analytical modules, each focusing on a specific dimension of the problem and update synchronously based on user interactions.

7.3.1 National Overview Dashboard

This module provides a high-level summary of electricity demand across the system. It includes KPI cards for total demand, average demand, shortage, and growth metrics, along with a time-series visualization of demand trends and rolling averages. A comparative bar chart highlights top contributing states.

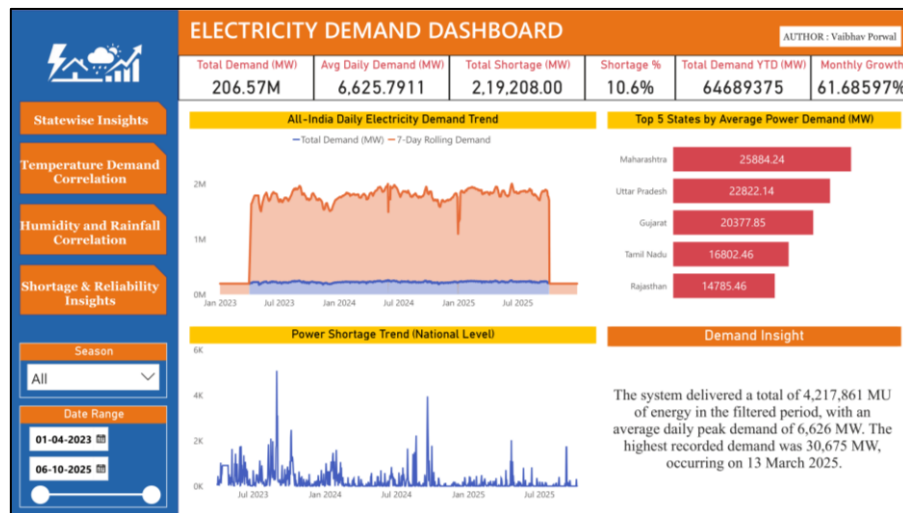


Figure 7.2: National Overview Dashboard

7.3.2 Statewise Insights Dashboard

This module focuses on regional analysis by comparing demand, shortage, and weather patterns across states. It includes combined charts such as demand vs shortage, demand vs rainfall, and demand vs temperature. These visualizations enable identification of regional disparities and trends, supporting targeted analysis of high-demand or high-risk regions.

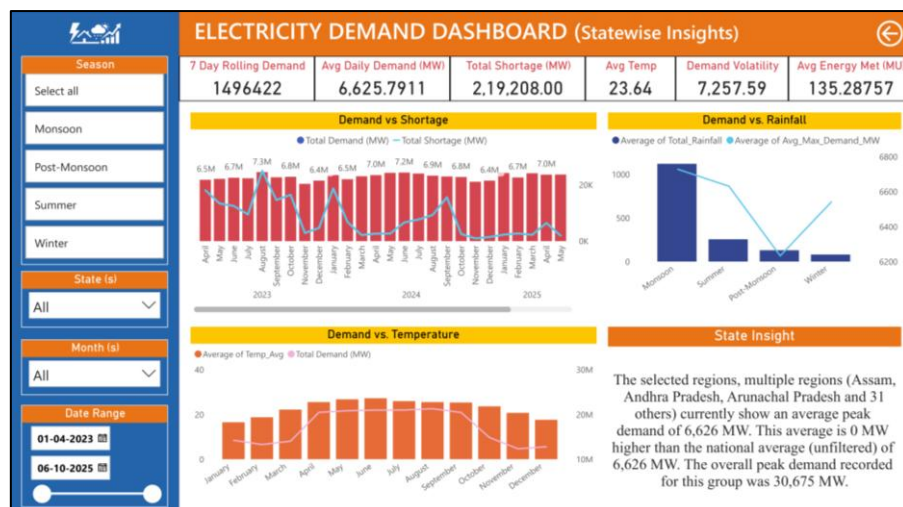


Figure 7.3: Statewise Insights Dashboard

7.3.3 Temperature–Demand Correlation Dashboard

This module analyzes the relationship between temperature and electricity demand using scatter plots and correlation rankings. Scatter visualization highlights distribution patterns across states, while ranking charts identify regions with strongest and weakest correlations. This helps in understanding climate sensitivity and its impact on demand.

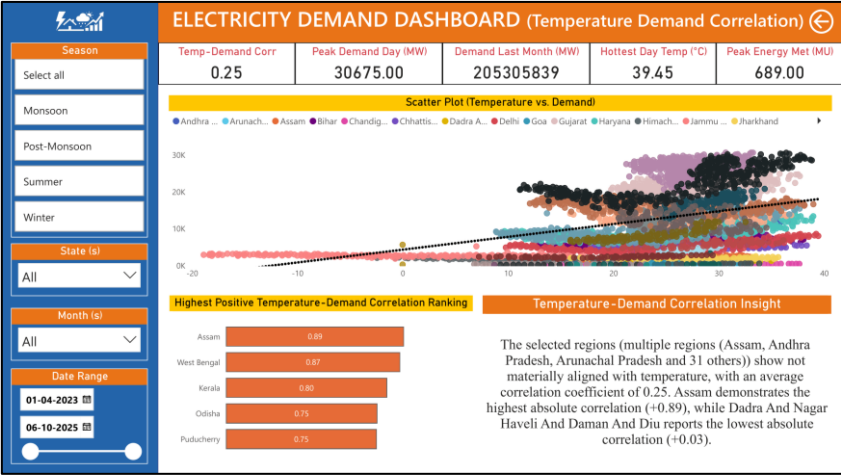


Figure 7.4: Temperature–Demand Correlation Dashboard

7.3.4 Humidity and Rainfall Correlation Dashboard

This module extends correlation analysis to humidity and rainfall variables. Scatter plots combined with trendlines illustrate how demand varies with atmospheric conditions. The module supports comparative interpretation across multiple weather parameters, enabling a broader understanding of environmental influence on electricity consumption.

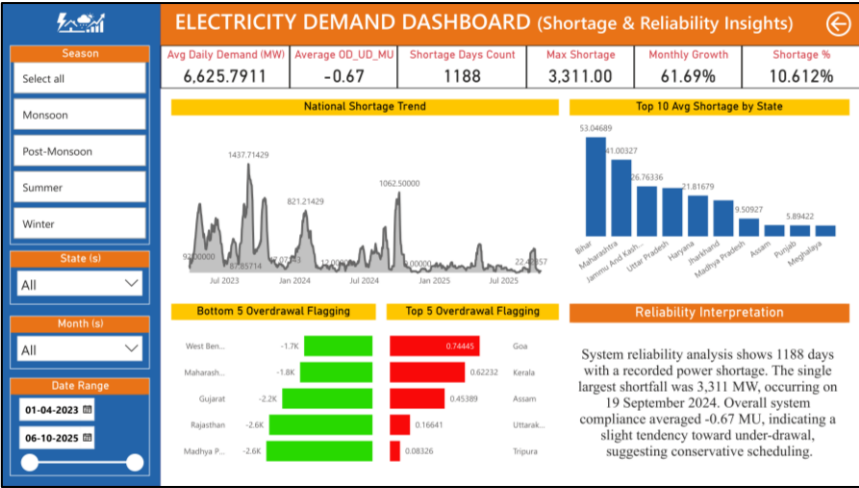


Figure 7.5: Humidity & Rainfall Correlation Dashboard

7.3.5 Shortage and Reliability Insights Dashboard

This module evaluates system reliability through shortage trends and operational indicators. It includes time-series visualization of shortages, state-level shortage comparisons, and overdrawal/underdrawal analysis. These visuals provide insights into grid stress, inefficiencies, and reliability risks, supporting operational decision-making.

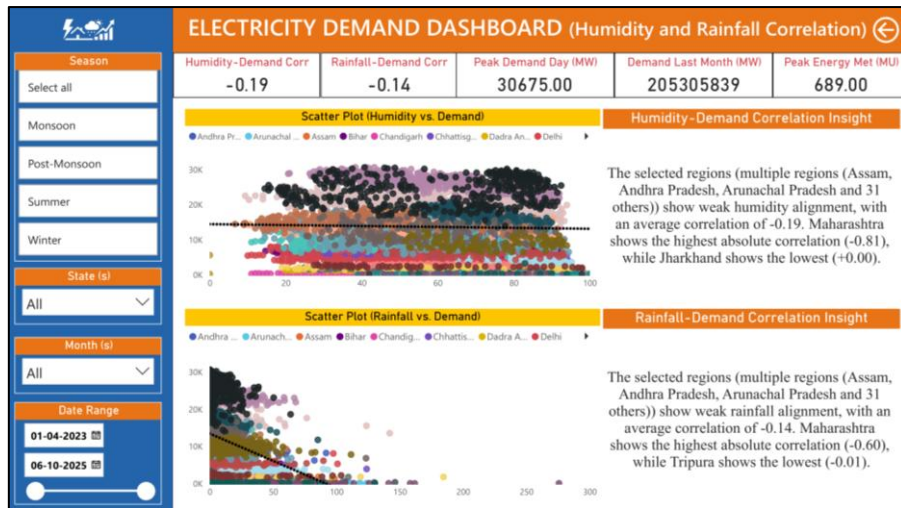


Figure 7.6: Shortage & Reliability Dashboard

7.4 Deployment on Power BI Service and Microsoft Fabric

The dashboard is deployed on the Power BI Service, enabling centralized cloud access, sharing, and automated refresh capabilities. Integrated with Microsoft Fabric, it operates within a unified ecosystem that supports scalable data management and seamless report consumption. Users can subscribe to dashboard views and receive automated email snapshots and mobile notifications, ensuring continuous monitoring of key metrics without manual access.

The dashboard is responsive and also supports *portrait mode on mobile devices*, ensuring accessibility and usability across platforms. Additionally, dynamic narrative measures implemented using DAX generate context-aware textual insights that adapt to user selections, enhancing interpretability alongside visual analytics.

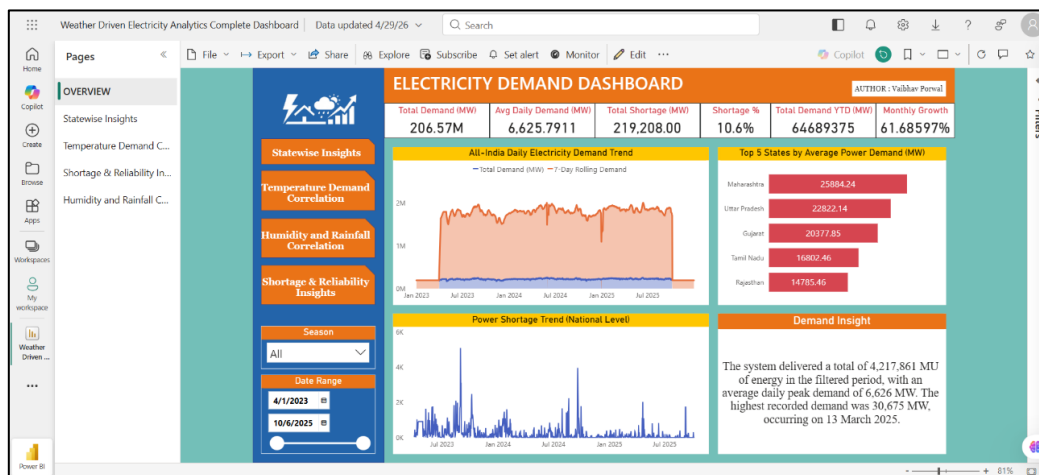


Figure 7.7: Power BI Service Dashboard View (Web Interface)

Chapter 8 – Backend Architecture and API Design

8.1 Architectural Design of Backend System

The backend architecture is designed as a modular, service-oriented system built using FastAPI, ensuring high-performance request handling, scalability, and seamless integration with machine learning pipelines. The architecture follows a layered approach comprising API routing, business logic processing, model inference, and database interaction. This includes:

- Asynchronous request handling using FastAPI
- Strict data validation via Pydantic schemas
- Decoupled machine learning inference layer
- Persistent storage through Prisma ORM and SQLite
- Secure authentication with role-based access control

This modularization ensures maintainability, extensibility, and efficient integration with frontend interfaces and analytical dashboards.

8.2 Backend Layered Architecture

The backend is structured into distinct logical layers to streamline execution flow and maintain separation of concerns.

- API Layer:
 - Handles HTTP requests and routing
 - Implements RESTful endpoints for prediction and analytics
- Service Layer:
 - Contains business logic for prediction and shortage computation
 - Orchestrates preprocessing and model execution
- Model Layer:
 - Executes trained machine learning models
 - Generates demand forecasts and shortage estimations
- Data Layer:
 - Manages persistent storage using Prisma ORM
 - Stores users, sessions, tokens, and system metadata

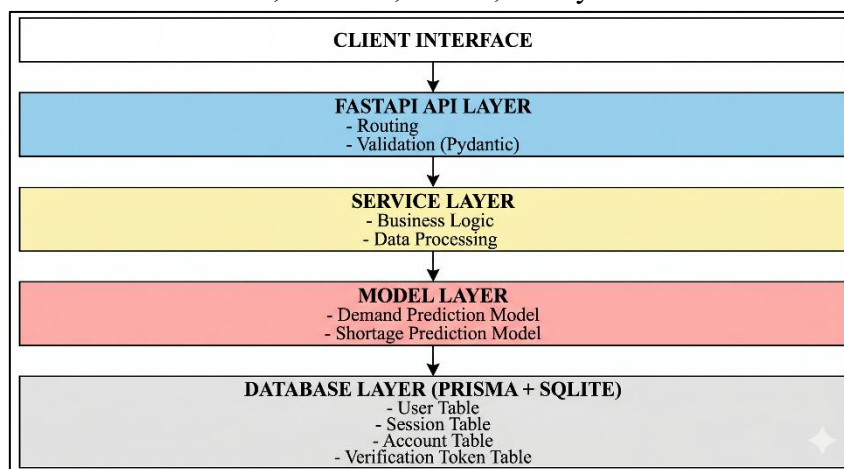


Figure 8.1: Unified FastAPI Backend Architecture with Database Integration

8.3 Database Architecture and Schema Design

The system utilizes Prisma ORM with SQLite to ensure structured, relational, and efficient data management. The database is designed to support authentication, authorization, and session management alongside system-level operations.

8.3.1 Core Entities

- User Model:
 - Stores user identity and credentials
 - Includes role-based access (VIEWER, ANALYST, ADMIN)
 - Maintains timestamps (createdAt, updatedAt)
- Account Model:
 - Supports OAuth providers
 - Stores access tokens and session metadata
 - Maintains provider-level uniqueness constraints
- Session Model:
 - Tracks active user sessions
 - Ensures secure session lifecycle management
- Verification Token Model:
 - Handles authentication validation workflows
 - Ensures secure token-based verification

8.3.2 Schema Design Features

- Use of cuid() for unique identifiers
- Cascading relationships for referential integrity
- Indexed and unique constraints for performance
- Role stored as string (SQLite compatibility)

This schema ensures secure user management and scalable authentication frameworks.

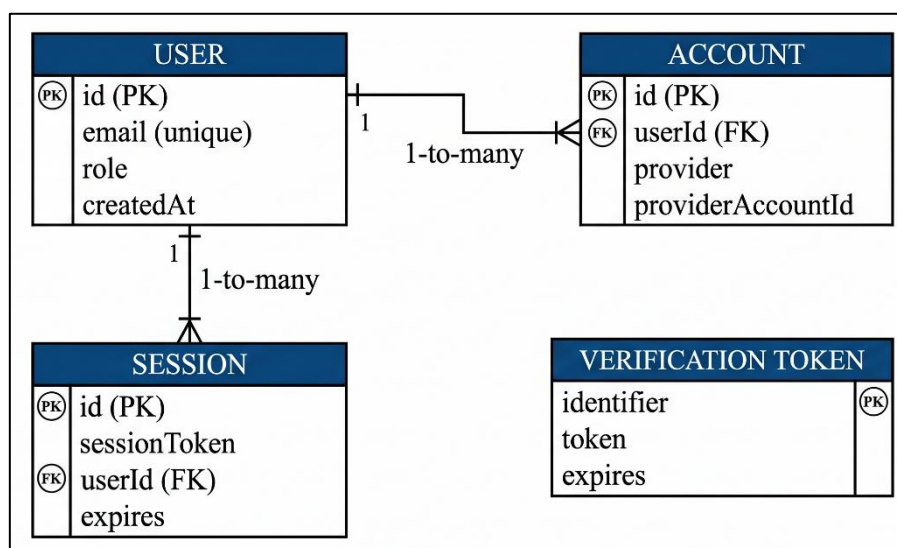


Figure 8.2: Relational Database Schema (Prisma-Based Design)

8.4 API Design and Endpoint Structure

The backend delivers RESTful APIs for frontend integration and analytics, built on stateless design, JSON-based communication, clear endpoint separation, and validation-first execution. Primary endpoints include:

- */predict-demand*: Inputs include State, Date, and Weather Data, while the output is predicted electricity demand.
- */predict-shortage*: Inputs include Demand forecast, weather inputs while Outputs are Shortage estimation and risk level.
- */auth/login, /auth/register*: Handles authentication and user creation.
- */admin/users*: Role management and user control.

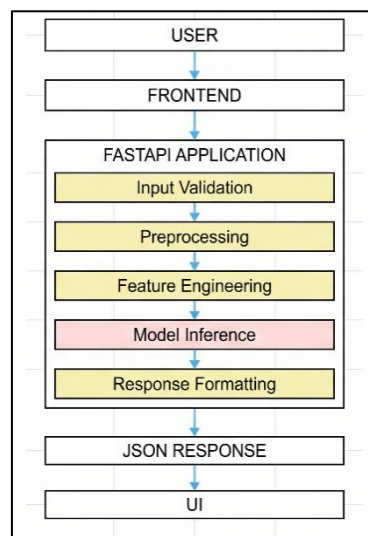


Figure 8.3: API Request-Response Lifecycle

8.5 Authentication and Role-Based Access Control

The system ensures secure authentication and authorization using session- and token-based mechanisms integrated with Prisma. Role tiers include Viewer (read-only access), Analyst (prediction and analytics access), and Admin (full system control with user management). Core security features encompass backend credential validation, role-based access control, secure session handling, and token-based verification.

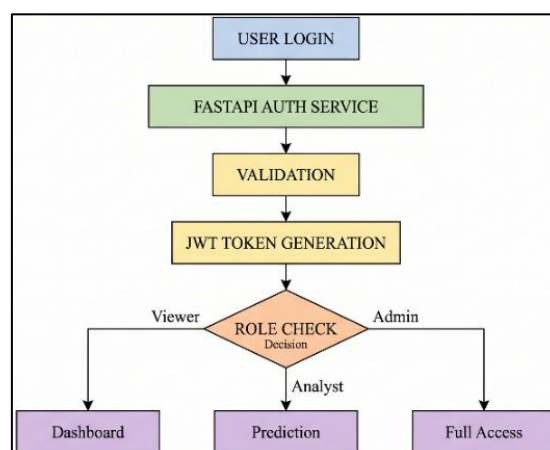


Figure 8.4: Authentication and Role-Based Access Control Flow

Chapter 9 – Interface Architecture and Deployment

9.1 Interface Design Language and Visual System

The frontend adopts a gradient-driven, modular card-based design to ensure clarity, consistency, and analytical readability. It emphasizes structured hierarchy, minimal clutter, and seamless navigation through a unified visual language.

A standardized color system governs the interface: primary blue (#2364AA) for structure, secondary blue (#3DA5D9) and teal (#73BFB8) for gradients, orange (#EA7317) for key actions, and yellow (#FEC601) for alerts, ensuring visual coherence and functional distinction. Typography, contrast, and spacing are optimized for readability, while soft shadows enhance component separation across inputs, outputs, and analytics.

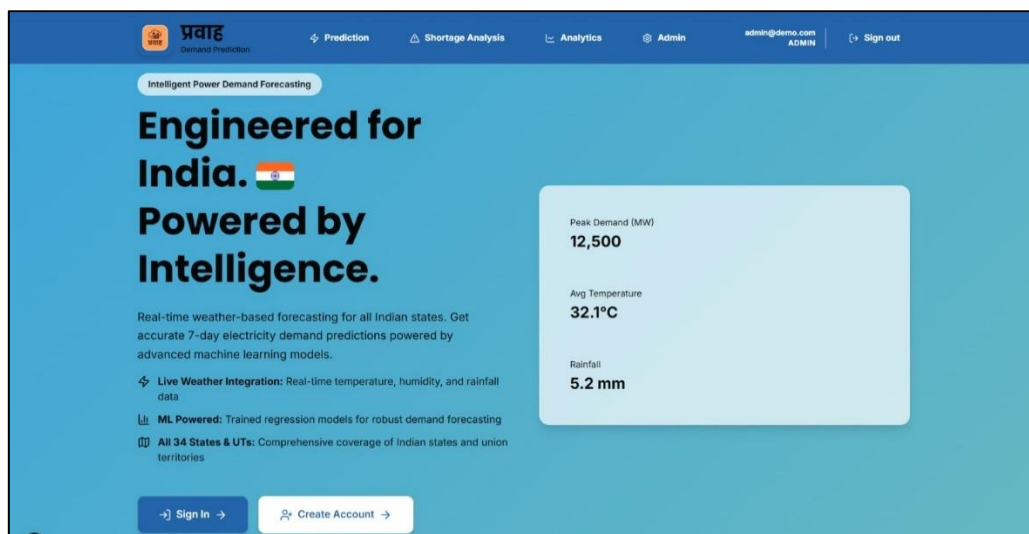


Figure 9.1: Prawah Interface Landing Page

9.2 Navigation Architecture and Layout Composition

The interface follows a top-navigation architecture for seamless module transitions across Prediction, Shortage Analysis, Analytics, and Admin. It features a fixed header (#2364AA), a centralized content region, and a two-column grid with inputs on the left and output visualizations on the right ensuring discoverability and minimal interaction overhead.

9.3 Prediction Module Interface

The prediction module functions as the core interface for generating electricity demand forecasts from real-time inputs. The left panel hosts structured inputs (state, date) with contextual guidance, while a prominent action button (#EA7317) drives user interaction.

The right panel displays outputs—weather parameters, predicted demand (MW), confidence levels, and expected ranges—in a clear, structured format for quick interpretation. Low-latency response ensures a fast, seamless analytical experience.

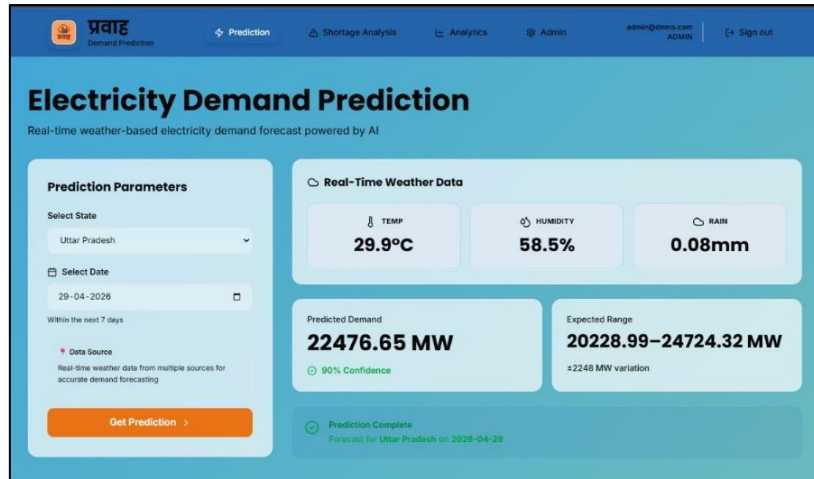


Figure 9.2: Electricity Demand Prediction Interface

9.4 Shortage Analysis Interface

The shortage analysis module is designed to provide risk-oriented insights, translating predictive outputs into actionable intelligence. The interface integrates multiple components, including risk probability, predicted shortage values, categorized risk levels, and supporting weather data. These elements are presented in a structured layout to maintain clarity and analytical coherence.

Color encoding plays a critical role in this module. High-risk conditions are emphasized using red tones, medium risk is indicated using #FEC601, and lower risk levels are represented through neutral or teal-based tones (#73BFB8). This visual differentiation enables immediate recognition of severity levels without requiring detailed interpretation.

Additionally, the recommendation panel converts analytical outputs into actionable suggestions, enhancing the system’s utility as a decision-support tool. The integration of prediction, visualization, and recommendation within an interface strengthens its applicability.

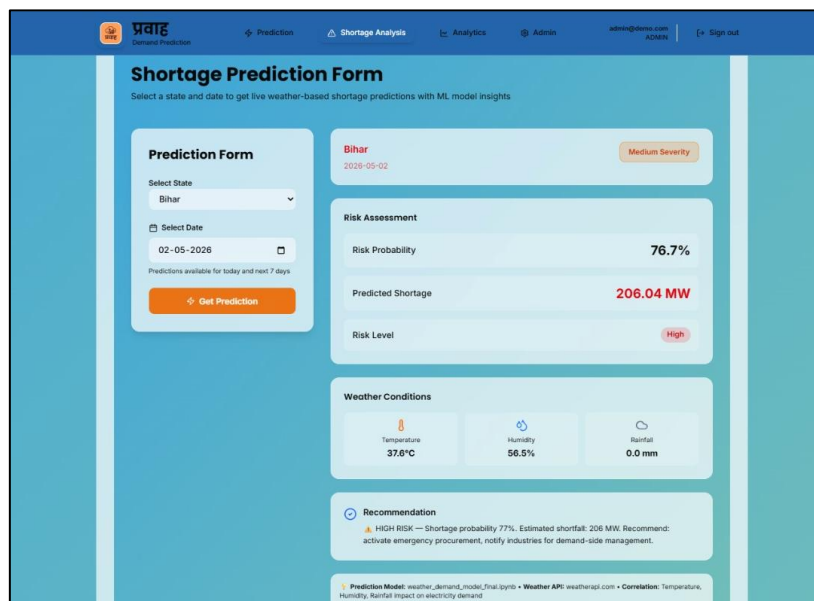


Figure 9.3: High Risk Shortage Prediction

9.5 Analytics Interface – PowerBI Dashboard

The analytics interface follows the shortage module, forming a flow from prediction → shortage → historical analysis. A Microsoft Power BI dashboard is embedded via secure iframe, delivering trend analysis, shortage distribution, state comparisons, and demand–weather correlation from historical datasets. Aligned with the system palette (#73BFB8, #2364AA → #3DA5D9 → #73BFB8), it enables a hybrid predictive-analytical platform combining real-time forecasts with historical insights. Users can subscribe to dashboards for automated email snapshots and mobile notifications, ensuring continuous monitoring.

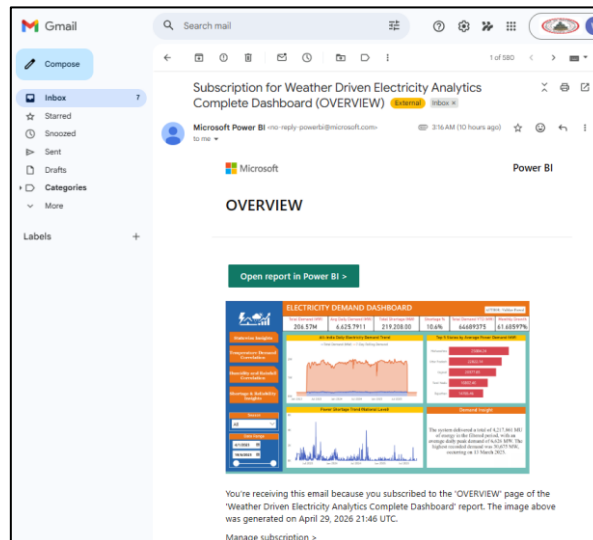


Figure 9.4: Subscription Email Notification (Overview Page Snapshot)

9.6 Authentication Interface

The authentication system prioritizes usability with secure design. The login interface includes email and password fields, a prominent sign-in button (#EA7317), and a registration link for onboarding. The registration interface captures name, email, and password with confirmation, using a minimal layout to ensure efficient and error-free user interaction.

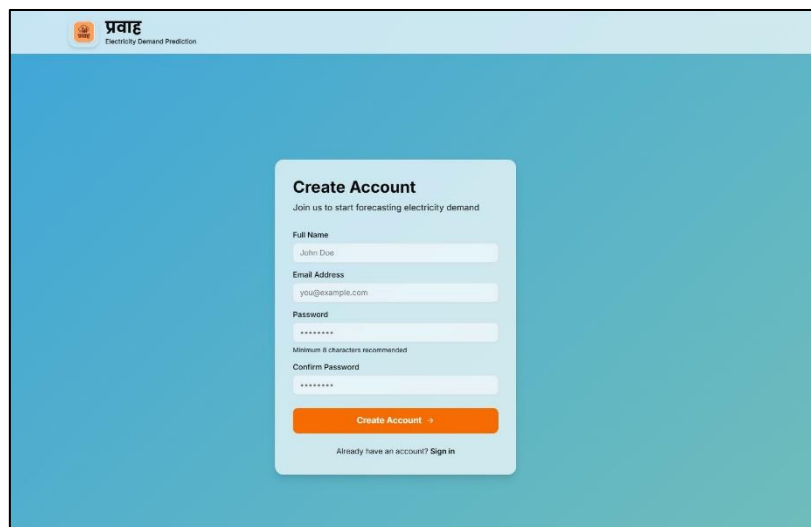


Figure 9.5: Registration Screen Screenshot

9.7 Admin Dashboard and Role Management

The admin dashboard centralizes user and access management with summary metrics and a structured user table. Role updates can be performed directly within the interface for operational efficiency.

Access is tiered as viewer (read-only), analyst (prediction and analytics), and admin (full control), with clear visual labels to streamline user management and reduce complexity.

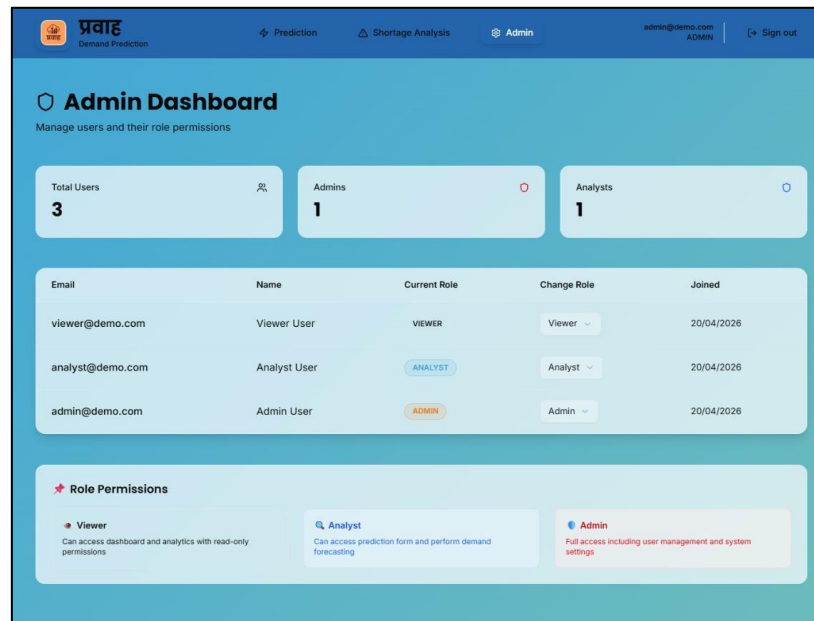


Figure 9.6: Administrative Role Management Interface

9.8 Frontend–Backend Interaction

The frontend interacts with backend services via RESTful APIs, where user actions trigger requests processed by FastAPI, invoking ML models as needed. Responses are returned in JSON and rendered dynamically. A stateless architecture ensures scalability and reliability, while low-latency processing supports seamless, real-time analytical workflows.

The frontend implementation achieves a balanced integration of design and functionality. The use of a consistent HEX-based color system ensures visual coherence, while the structured layout enhances usability and interpretability. The interface effectively supports real-time prediction, risk analysis, and administrative control, establishing a robust and user-centric analytical platform.

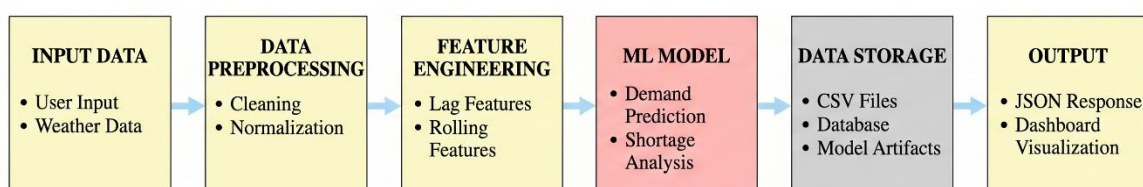


Figure 9.7: Data Flow and Processing Pipeline

Chapter 10 – Testing and Evaluation

10.1 Data Processing and Validation Testing

The integrity of the forecasting pipeline is contingent upon robust preprocessing and temporal consistency. Validation was conducted to ensure correctness, completeness, and transformation fidelity.

TABLE 10.1: DATA PROCESSING TEST CASES

Test Case ID	Scenario	Input	Expected Outcome	Status
DP-01	Missing values handling	Null weather entries	Imputation using interpolation/mean	Pass
DP-02	Date parsing validation	Mixed date formats	Standardized datetime format	Pass
DP-03	Feature scaling	Raw numerical values	Normalized/scaled dataset	Pass
DP-04	Outlier handling	Extreme temperature values	Capped/retained based on threshold	Pass
DP-05	Time ordering	Unsorted timestamps	Chronologically ordered dataset	Pass

10.2 Time-Series Integrity and Data Leakage Prevention

The system enforces strict temporal validation to eliminate forward-looking bias.

- TimeSeriesSplit methodology applied
- Training restricted to past observations only
- Validation performed on unseen future windows

TABLE 10.2: TIME-SERIES VALIDATION TESTING

Test Case ID	Scenario	Expected Behavior	Status
TS-01	Forward chaining validation	No future data leakage	Pass
TS-02	Rolling window training	Progressive model update	Pass
TS-03	Temporal consistency	Chronological integrity maintained	Pass

10.3 Machine Learning Model Evaluation

The predictive performance of models was evaluated using standard regression metrics. Multiple models were benchmarked to ensure optimal selection. The ensemble model demonstrates superior predictive capability, balancing variance and bias effectively across temporal patterns.

TABLE 10.3: MODEL PERFORMANCE COMPARISON

Model	RMSE	MAE	R ² Score
Linear Regression	1850.42	1423.11	0.87
XGBoost Regressor	1324.76	1012.58	0.92
Prophet	1498.33	1187.90	0.89
Ensemble Model	1210.55	945.21	0.94

10.4 Backend API Testing

Backend services were evaluated for correctness, response efficiency, and robustness under varied conditions.

TABLE 10.4: API FUNCTIONAL TESTING

Test Case ID	Endpoint	Scenario	Expected Response	Status
API-01	/predict	Valid input	Demand prediction output	Pass
API-02	/predict	Missing parameters	Validation error (422)	Pass
API-03	/weather	External API fetch	Real-time weather data	Pass
API-04	/admin/users	Role access	Authorized access only	Pass

TABLE 10.5: API PERFORMANCE TESTING

Scenario	Avg Response Time	Throughput	Status
Single request	~120 ms	—	Pass
50 concurrent users	~280 ms	180 req/sec	Pass
Invalid request handling	<100 ms	—	Pass

10.5 Database and ORM Testing

Database integrity and relational consistency were validated using Prisma schemas.

TABLE 10.6: DATABASE TESTING

Test Case ID	Scenario	Expected Outcome	Status
DB-01	User creation	Record stored correctly	Pass
DB-02	Role assignment	Correct role mapping	Pass
DB-03	Session handling	Token stored and validated	Pass
DB-04	Cascade deletion	Related records removed	Pass
DB-05	Unique constraints	Duplicate prevention	Pass

10.6 Frontend Interface Testing

The user interface validation strategy extends to ensuring adaptive responsiveness across devices and screen resolutions, maintaining seamless interaction flows under varying load conditions.

TABLE 10.7: UI FUNCTIONAL AND PERFORMANCE TESTING

Test Case ID	Component	Scenario	Expected Behaviour	Status
UI-01	Prediction form	Valid input	Displays prediction results	Pass
UI-02	Input validation	Empty fields	Warning prompt	Pass
UI-03	Dashboard cards	API response	Dynamic updates	Pass
UI-04	Navigation	Page switching	Seamless transition	Pass
UI-05	-	Page Load Time	< 2 seconds	Pass
UI-06	-	API data rendering	Real-time update	Pass
UI-07	-	Mobile responsiveness	Mobile responsiveness	Pass

10.7 Power BI Dashboard Validation

The analytical dashboard was evaluated for correctness using historical datasets.

TABLE 10.8: POWER BI TESTING

Test Case ID	Scenario	Expected Outcome	Status
BI-01	Data loading	Historical data displayed	Pass
BI-02	Visualization accuracy	Correct aggregation	Pass
BI-03	Filtering	Dynamic filtering applied	Pass
BI-04	Trend analysis	Seasonal patterns visible	Pass

10.8 Edge Case and Robustness Testing

The system was stress-tested against abnormal and boundary conditions.

TABLE 10.9: EDGE CASE TESTING

Test Case ID	Scenario	Input	Expected Behaviour	Status
EC-01	Extreme temperature	50°C	Demand spike prediction	Pass
EC-02	Missing weather data	Null values	Graceful fallback	Pass
EC-03	Invalid state	Unknown input	Error response	Pass
EC-04	Future unseen date	Beyond dataset	Stable prediction	Pass

10.9 System Integration Testing

End-to-end workflow validation ensured seamless interaction between components.

TABLE 10.10: INTEGRATION TESTING

Test Case ID	Scenario	Input	Expected Behaviour	Status
INT-01	Extreme temperature	50°C	Demand spike prediction	Pass
INT-02	Missing weather data	Null values	Graceful fallback	Pass
INT-03	Invalid state	Unknown input	Error response	Pass
INT-04	Future unseen date	Beyond dataset	Stable prediction	Pass

10.10 System Reliability and Failure Handling

Resilience testing validated system behaviour under failure scenarios.

TABLE 10.11: RELIABILITY TESTING

Scenario	Expected Behaviour	Status
API Failure	Error message displayed	Pass
Database disconnection	Retry mechanism	Pass
External API timeout	Fallback handling	Pass

Chapter 11 – Results and Discussion

11.1 Data Engineering and Preprocessing Outcomes

- Integration of 35,000+ grid records and 33,121 weather records into a unified dataset ensured complete temporal coverage (Apr 2023 – Oct 2025), enabling robust longitudinal analysis across all states.
- Correction of 1,443 zero-valued demand entries using forward–backward interpolation removed artificial dips while preserving seasonal patterns, ensuring statistical validity of time-series modelling.
- Standardization of ~200 state name inconsistencies enabled accurate joins, eliminating entity fragmentation and ensuring state-level analytical integrity.
- Coordinate-weighted aggregation of weather data improved geospatial representativeness, ensuring that climatic signals reflect true regional variability rather than point-based bias.
- Final dataset with 16 engineered features (including shortage indicators and temporal encodings) enabled multi-dimensional modeling across demand, climate, and stress.

11.2 Weather – Power Demand Analytical Insights

- National-level correlations (Temperature: 0.286, Humidity: –0.188, Rainfall: –0.113) are statistically significant yet weak, demonstrating that aggregation masks regional climatic effects (averaging fallacy).
- *19 out of 34 states (~56%) exhibit $|r| > 0.5$* , proving strong regional climate sensitivity and validating the need for state-level modeling instead of centralized forecasting.
- High positive temperature correlations in states like Assam (0.885), West Bengal (0.866), Kerala (0.801) confirm cooling-driven demand behavior, particularly in humid and coastal regions.
- Negative correlations in states such as Madhya Pradesh (–0.587), Jammu & Kashmir (–0.573) indicate inverse or structurally driven demand patterns, reflecting heating demand or industrial load variations.
- Humidity exhibits dual behavior (positive in eastern states, strongly negative in southern states), indicating complex weather interactions affecting consumption.
- Rainfall shows weak and inconsistent correlations, confirming limited direct predictive value, but contributes when combined in composite modelling.

11.3 Weather-Temporal-Shortage Dynamics

- *Composite Weather Index (CWI) impact:* Despite weak national correlation, CWI demonstrates enhanced explanatory strength at state-level, validating multi-variable climatic representation for demand modelling.
- *Temporal stability extraction:* Rolling averages (7-day, 14-day) effectively reveal seasonal demand cycles and peak load windows, improving forecast consistency and noise reduction.
- *Extreme temperature influence:* Demand behavior shows clear escalation beyond 35 °C thresholds, confirming temperature as a direct trigger for peak load conditions.

- *Shortage concentration pattern*: Deficits are regionally clustered rather than uniformly distributed, with specific states consistently exhibiting higher shortages, indicating localized grid stress zones.
- *Operational inference*: The interaction of climatic indices, temporal trends, and shortage localization establishes the need for integrated, time-aware, and region-adaptive forecasting frameworks.

11.4 Clustering and Regional Segmentation Insights

- Ward's hierarchical clustering produced *four statistically stable clusters* ($CCC = 0.802$), validating strong structural grouping in the dataset.
- Identified clusters represent distinct demand–climate regimes:
 - Coastal humid regions → climate-driven cooling demand
 - Northern plains → high temperature elasticity
 - Hilly regions → inverse demand behavior
 - Industrial regions → relatively stable demand
- Clusters exhibit clear geographical coherence, confirming that electricity demand patterns are spatially structured and not random.
- Segmentation reduces intra-group variance, improving model generalization and forecasting precision.

11.5 Machine Learning Results and Interpretation

- *Demand prediction achieves very high accuracy* ($R^2 = 0.9936$; $RMSE \approx 523$ MW), demonstrating that electricity demand is predominantly governed by temporal patterns (lags, seasonality) and weather features, enabling highly reliable short-term forecasting.
- Linear Regression emerges as the best-performing model, indicating that after feature engineering, relationships become largely linear, minimizing need for complex models.
- Tree-based models (Random Forest, XGBoost) provide robustness but lower accuracy, suggesting limited non-linear influence compared to dominant temporal trends.
- SVR performs poorly ($R^2 = 0.3294$; $RMSE > 5500$ MW), confirming that models without effective temporal representation are unsuitable for demand forecasting.
- Prophet-based forecasting shows weak predictive accuracy (Mean $R^2 < 0$) but effectively captures trend, seasonality, and uncertainty, making it suitable for planning rather than precise prediction.
- Shortage prediction demonstrates strong discriminative capability (ROC-AUC = 0.9069), validating effective identification of high-risk scenarios despite severe class imbalance.
- High accuracy (96%) is inflated due to imbalance (~3.8% shortage cases), therefore, recall and ROC-AUC are more reliable indicators of performance.
- Low recall (0.24) reflects a conservative prediction strategy, prioritizing precision and reducing false positives, but indicating partial under-detection of shortage events.
- Shortage magnitude prediction achieves moderate accuracy ($R^2 \approx 0.50$; $MAE \approx 98$ MW), which is operationally acceptable given the stochastic nature of shortages.
- *Two-stage modeling (classification + regression)* improves system reliability, effectively handling zero-inflated data and separating occurrence from severity.

TABLE 11.1: SHORTAGE PREDICTION PERFORMANCE

Metric	Value
Accuracy	96%
Precision	0.52
Recall	0.24
F1-Score	0.33
ROC-AUC	0.9069

11.6 System Integration and Operational Outcome

- The end-to-end pipeline (data ingestion → preprocessing → ML inference → FastAPI → frontend → Power BI) executes seamlessly, confirming complete system coherence and production readiness.
- FastAPI backend maintains low-latency responses (~120–280 ms), ensuring real-time prediction capability suitable for operational deployment.
- Modular architecture (API → service → model → database) enables scalable system design, allowing independent upgrades without affecting overall stability.
- Database layer (Prisma + SQLite) ensures secure data persistence and role-based access control, supporting reliable multi-user interaction.
- Frontend interface delivers real-time visualization of predictions and analytics, enabling intuitive interaction and reducing complexity for end users.
- Embedded Power BI dashboard integrates historical analytics with predictions, enabling combined descriptive and predictive decision support.
- Deployment on Render and Vercel ensures high availability and accessibility, validating the system as a fully functional web-based solution.

11.7 System Deliverables

1. *Demand Forecasting Interface*: Primary user-facing module delivering state-wise demand predictions in real-time, forming the core functional output of the system.
2. *Analytics Dashboard (Power BI)*: Interactive visualization layer providing historical trends, correlation insights, and narrative-driven analytics to support decision-making.
3. *Shortage Prediction Engine*: Risk assessment module estimating probability and magnitude of power shortages, enabling proactive grid planning.
4. *Prediction APIs (FastAPI)*: Scalable backend endpoints powering forecasting and shortage modules, ensuring seamless integration and system interoperability.
5. *Curated Analytical Dataset*: Cleaned and feature-engineered dataset integrating demand and weather variables, serving as foundation for modeling and analytics.
6. *Deployed Web Application*: Cloud-hosted system (Render for backend, Vercel for frontend) ensuring real-time accessibility and scalability in a production environment.

Chapter 12 – Conclusion and Future Work

12.1 Conclusion

The project presents a comprehensive and fully integrated system for electricity demand forecasting and shortage prediction, combining data engineering, analytical modeling, machine learning, backend services, and interactive visualization into a unified framework. The results demonstrate that electricity demand is strongly governed by temporal patterns and climatic variables, with significant regional variability necessitating state-level modeling approaches. The machine learning framework achieves high predictive reliability, while the incorporation of shortage prediction extends the system from pure forecasting to actionable decision support.

The seamless integration of predictive models with a web-based interface and analytical dashboard ensures both usability and interpretability. Furthermore, successful cloud deployment validates the system's scalability, accessibility, and readiness for real-world applications in energy planning and grid management.

12.2 Future Scope

- Integration of advanced deep learning models such as LSTM and Transformer architectures for improved temporal modelling.
- Development of real-time data pipelines for continuous learning and live forecasting.
- Extension to district-level or sub-regional forecasting for finer spatial granularity.
- Incorporation of additional factors such as economic and industrial indices.
- Implementation of automated alert and recommendation engines for grid management.
- Expansion into mobile-first and offline-capable platforms for broader accessibility.

12.3 Limitations

The system, while robust and effective, has certain limitations that influence its scope of applicability. Model performance may vary under extreme or previously unseen climatic conditions, affecting generalization in rare scenarios. Dependence on external weather data sources introduces potential variability and latency in real-time predictions.

The use of state-level aggregated data restricts finer spatial analysis and may overlook localized demand variations. Additionally, shortage prediction is influenced by class imbalance and external operational factors such as policy decisions and supply constraints, which are not fully captured within the current modeling framework.

12.4 Closing Remarks

The work successfully demonstrates how the integration of data analytics, machine learning, and scalable system design can address complex real-world problems in the energy domain. By bridging predictive modeling with deployment and visualization, it evolves into a practical decision-support system rather than a purely academic exercise.

The framework established in this work is extensible and adaptable, providing a strong foundation for future research and development in intelligent energy forecasting and smart grid management systems.

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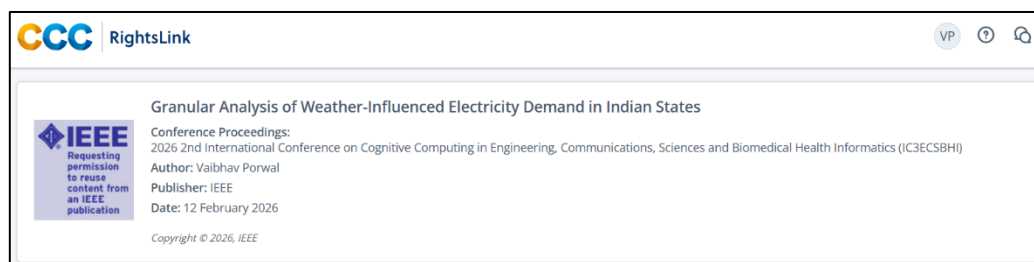


Figure A.1: IEEE RightsLink Authorization

Appendix – B (Excerpt from Published Paper)

Granular Analysis of Weather-Influenced Electricity Demand in Indian States

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Abstract—While when the demand for electricity is estimated in India at the national level, the difference in which some regions are affected by weather to a greater or lesser extent is obscured by averaging the data. By using daily data from April 2023 to October 2025, this study aims to examine the effect of weather on energy demand at the state level by using Grid India's Power Supply Position reports in association with Open-Meteo archives by using a structured data preparation methodology. The results show considerable variation in geographic distribution, with more than half of the states showing high local dependency, whereas the relationship between temperature and demand at the national level remains weak. At some locations in northern and eastern states, high-temperature days increase peak demand by 3 to 6 gigawatts, showing that reserve planning should be region-specific. Even though Bihar and Maharashtra are identified as stressed regions by performing a scarcity analysis, using a composite weather indicator that includes temperature, humidity, and rainfall improves the accuracy of state-level relationships. The results show that treating the national grid as a single entity obscures operational risks and highlights the need for flexible weather-informed planning for India's energy security.

Index Terms—Electricity demand, weather sensitivity, regional analysis, grid resilience, India.

I. INTRODUCTION

The demand of electricity in India is one of the most complex consumption landscapes in the world—an interconnected system that serves over 1.4 billion people across extremely different climates, economic profiles, and infrastructure capacities. Maintaining reliability in such an interconnected system requires constant adaptation to weather changes that occur in the short term as well as the long term with respect to climatic changes. However, the analytical tools that guide national-level electricity planning tend to rely on aggregated data that doesn't suit such diversity and instead aggregates all into one category. When the national weather and demand metrics are merged into one category, the level of correlation is low, indicating

that temperature and electricity usage are noncoherent with each other.

The weak correlations are because of the scale anomalies. Averaging leads to the 'averaging fallacy,' where the lack of sensitivity in demand to weather in certain states and the severe sensitivity in others are not reflected. This has major implications in the estimation of the risk in high-sensitivity states and the estimation of reserves in low-sensitivity states. The purpose of this research is to reinvestigate the relationship between weather and demand in India at the state level. It makes use of the 917 consecutive days of operational data provided by Grid India and the parallel weather data to correlate the demand and weather. It also attempts to bring out the heterogeneity in the averages. It does this by not only employing the correlation analysis but also by finding the states where there are persistent shortages during thermal stress, the seasonal demand cycles in the states where there are shortages during thermal stress, and the composite weather index.

The structure of the remaining part of the paper is as follows. In Section II, the related works on the weather-dependent electricity demand are presented. In Section III, the dataset, preprocessing pipeline, and feature generation procedure are explained. The methodology is explained in Section IV. The main results are presented in Section V, and the limitations are discussed in Section VI. The conclusion is presented in Section VII, which summarises the insights, implications, and ways forward.

II. RELATED WORK

Reliable data collection is necessary to understand the impact of weather on electricity usage [2]. In the past, the study on the impact of weather on electricity demand in India was based on the combined data from the entire country [3], which does not reveal the impact of weather on different states.

Executive Summary

This project presents a complete end-to-end system designed to understand and predict electricity demand across Indian states, along with identifying potential power shortages. The work is based on the idea that electricity consumption is influenced by time patterns and weather conditions such as temperature, humidity, and rainfall. By combining these factors into a structured dataset, the project establishes a strong foundation for meaningful analysis and prediction.

The system follows a clear and logical flow. It begins with data preparation and feature engineering, where important patterns such as past demand through lag features and short-term trends through rolling averages are captured. Based on this, machine learning models are developed to predict future demand with high accuracy. In addition to forecasting, a separate approach is used to detect shortages by first identifying whether a shortage will occur and then estimating its magnitude. This two-step method improves the reliability and practical usability of the predictions.

To ensure accessibility, a web-based interface is developed that allows users to input parameters and obtain predictions in real-time. Alongside this, an interactive dashboard provides insights into historical trends and relationships between weather and demand, enabling better understanding of system behaviour.

The system is deployed online, making it accessible and suitable for real-world use. Overall, the project demonstrates how data, machine learning, and visualization can be integrated to build a practical decision-support tool. It highlights the importance of regional-level analysis and shows how technology can contribute to improved planning and management of electricity demand.